FastSAM-Guided Adversarial Learning for Unsupervised Multimodal Remote Sensing Change Detection

Zhifu Zhu, Xiping Yuan, Shu Gan, Weidong Luo, Cheng Chen, and Xuan Li

***Abstract*—Multimodal change detection (CD), due to its ability to flexibly adapt to data acquired from different types of sensors, has become an important research direction in the field of remote sensing. However, existing methods generally lack feature representations with sufficient generalization capacity, leading to pronounced performance degradation when applied across diverse scenes. To address this limitation, we propose a novel FastSAM-guided adversarial learning (FastSAM-GAL) framework for unsupervised multimodal CD. The proposed framework effectively mitigates cross-modal discrepancies via adversarial learning and constructs a dual-branch architecture comprising a FastSAM encoder and a remote sensing encoder, thereby fully exploiting the complementary information between generic visual semantic features and domain-specific remote sensing representations to substantially enhance cross-scene generalization capability. Furthermore, a progressive multiscale perception adaptor (PMSPA) and a spatial-frequency collaborative edge augmentation (SFCEA) module are specifically designed to further improve FastSAM-GAL's ability to perceive objects at different scales and capture change boundaries. Finally, comparative experiments on five real-world multimodal CD datasets against 12 state-of-the-art methods demonstrate that the proposed FastSAM-GAL maintains stable and reliable detection performance across multimodal remote sensing data with diverse scenes, with its κ metric significantly outperforming all competing methods.**

***Index Terms*—Image translation, multimodal change detection (CD), remote sensing, vision foundation model.**

I. Introduction

REMOTE sensing image change detection (CD), as an indispensable component of Earth observation tasks, aims to identify changes in land cover by comparing two or more images acquired by sensors over the same geographic area at different time phases [1]. With the rapid advancement of satellite technology, CD has progressively permeated multiple domains including disaster assessment [2], ecological environment monitoring [3], and urban expansion analysis [4], playing a pivotal role in evidence-based decision-making and emergency response.

Depending on the source of sensor data employed, CD can generally be categorized into two broad classes: unimodal (homogeneous) CD and multimodal (heterogeneous) CD. Over the past decades, unimodal CD has been extensively studied and widely applied, as the bitemporal images are acquired from the same sensor, allowing direct pixel-level or feature-level comparison for change extraction [5]. However, with the increasing complexity and diversity of real-world application scenarios, the limitations of unimodal CD have gradually become apparent. For instance, optical sensors are susceptible to atmospheric conditions such as cloud cover, haze, and illumination variations, making it difficult to acquire high-quality images under adverse weather conditions [6]. In contrast, although synthetic aperture radar (SAR) sensors possess all-weather, all-time imaging capabilities, their high cost and relatively limited data availability result in notable deficiencies in timely data acquisition for specific regions. The over-reliance on a single sensor significantly reduces the usability of unimodal CD under adverse conditions such as cloud obstruction or data absence, thereby severely limiting its application potential in emergency scenarios. To address these limitations, researchers have increasingly shifted their focus toward multimodal CD. Compared with unimodal CD, multimodal CD allows bitemporal images to be acquired from different types of sensors, substantially enhancing the flexibility and timeliness of data acquisition [7]. Taking earthquake disaster events as an example, optical images with rich spectral information can be utilized before the event. However, extreme weather conditions often occur after the event, preventing optical sensors from capturing surface information. In such cases, complementary SAR images can be employed as postdisaster observation data sources, enabling rapid assessment of the affected areas.

Although multimodal CD offers significant advantages, the challenges it faces should not be underestimated. Due to fundamental differences among sensors in terms of imaging mechanisms, radiometric characteristics, and noise distributions, the same object may exhibit drastically different visual appearances across multimodal images [8]. This cross-modal heterogeneity renders traditional change extraction methods based on direct comparison difficult to apply [9], and may even introduce a large number of pseudo-changes [see Fig. 1(a)].

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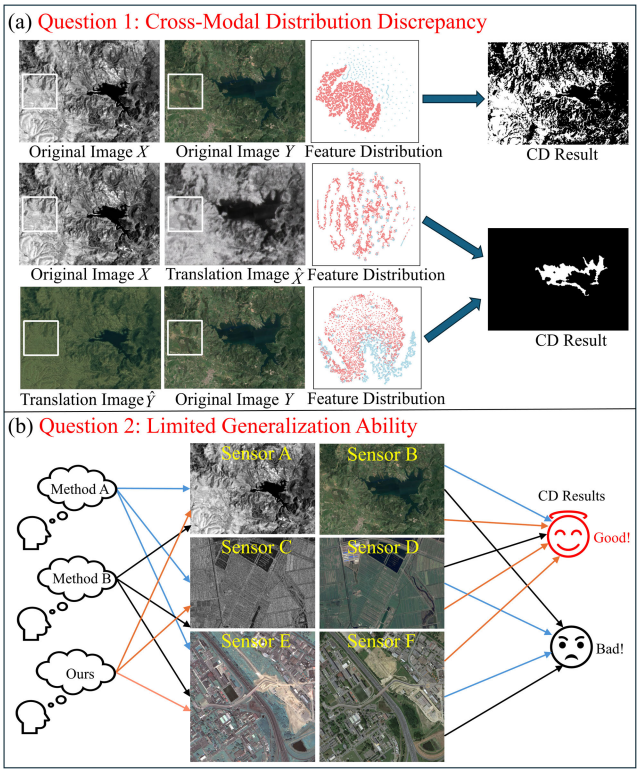


Fig. 1. Schematic of two key challenges in multimodal CD. (a) Direct comparison without mitigating modality discrepancies introduces numerous pseudo-changes. (b) Limited generalization of existing methods leads to strong performance on specific datasets but poor robustness across diverse scenarios.

Therefore, how to establish a bridge of comparability between images acquired from different sensors has become the primary task in multimodal CD. To achieve this goal, existing studies mainly explore two directions: traditional similarity measurement methods and deep-learning-based approaches. The former characterizes the similarity between multimodal images by designing specific handcrafted metrics, thereby constructing a comparable metric space. For example, Prendes et al. [10] proposed a framework based on multivariate statistical analysis, which utilizes mutual information and Copula theory to model the joint probability distribution between optical and SAR images, thereby extracting change information. Wan et al. [11] employed rank histograms to quantify the local statistical similarity between images from different sensors and measured changes by computing distances between histograms. Sun et al. [9] constructed graph structures for pre-event and postevent images, respectively, and extracted changes by comparing the structural differences between the graphs. Such methods heavily rely on the prior design of metric functions and struggle to capture deep semantic features, which limits their applicability in complex scenarios. In contrast, deep-learning-based approaches leverage the powerful feature learning capability of deep neural networks to construct cross-modal feature alignment or image translation models, thereby alleviating the limitations of traditional metric-based methods to some extent. Representative methods include the symmetric convolutional coupling network (SCCN) [12], conditional generative adversarial network (cGAN) [13], and code-aligned autoencoders (CAAs) [14].

Although the aforementioned methods have achieved notable success in multimodal CD tasks, certain limitations remain to be addressed. Specifically, existing approaches generally lack support from universal prior knowledge, and their feature representation capabilities are highly dependent on the distribution characteristics of specific datasets. When confronted with multimodal remote sensing data differing in sensor types, spatial resolutions, and change scenarios, these methods often struggle to maintain stable detection performance, with cross-scene generalization capability being significantly constrained [see Fig. 1(b)]. This issue is particularly pronounced under an unsupervised learning paradigm, where the absence of labeled information exacerbates the difficulty of generalization under limited data conditions, resulting in performance that can only be guaranteed in specific scenarios. To overcome this bottleneck, it is especially necessary to develop a multimodal CD framework with stronger cross-scene generalization capability.

In recent years, visual foundation models (VFMs), represented by the segment anything model (SAM) [15] and its derivative variants (FastSAM [16] and MobileSAM [17]), have rapidly advanced. Leveraging the rich general visual representations learned from large-scale pretraining on natural images, these models demonstrate strong cross-task and cross-scene generalization capabilities. Inspired by these advances, VFMs research has gradually expanded from the natural vision domain to the field of remote sensing. For example, Xiong et al. [18] proposed the dynamic one-for-all (DOFA) foundation model, which learns unified representations across data acquired from different sensors by constructing a wavelength-conditioned dynamic hypernetwork. Szwarcman et al. [19] developed Prithvi, a unified pretrained model trained on global-scale, multitemporal Earth observation data, demonstrating strong transferability across tasks such as land-cover mapping, disaster monitoring, and environmental assessment. HyperR3SNet [20] introduced self-supervised VFMs into remote sensing image analysis and designed a pixel-level hyperbolic-space loss function, effectively enhancing remote sensing image segmentation performance under limited annotation conditions. To address the challenges posed by the diversity of remote sensing data sources and significant seasonal variations, Li et al. [21] further developed a remote sensing VFM named SeaMo. By integrating temporal and modal information, SeaMo learns more robust and generalizable remote sensing representations. Subsequently, methods such as AutoSAM [22], Anchor-SAM [23], and SAM-SS [24] were proposed, further expanding the application potential of VFMs in remote sensing scene understanding and providing new solutions for the perception and interpretation of complex remote sensing scenarios.

Although the aforementioned VFMs achieve remarkable performance on remote sensing tasks, their complex architectures and substantial computational overhead inevitably compromise inference efficiency in practical deployments. In contrast, FastSAM adopts a lightweight architectural design that substantially reduces computational cost while retaining the powerful semantic representation capability of SAM.

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Therefore, we introduce FastSAM into unsupervised multimodal CD with the aim of leveraging its strong generalizable semantic priors to effectively compensate for the limited cross-scene generalization capability of existing methods.

Concretely, we propose a framework named FastSAM-guided adversarial learning (FastSAM-GAL). This framework employs FastSAM as a frozen encoder and combines it with a learnable remote sensing encoder to construct a dual-branch feature extraction architecture. The two branches work collaboratively to fully exploit the complementary information between general visual semantic features and remote sensing-specific domain features, thereby mitigating cross-modal heterogeneity while enhancing the model's cross-scene generalization capability. It is worth noting that, due to the inductive biases acquired from pretraining on natural images, FastSAM still exhibits certain limitations when applied to remote sensing images [25], such as limited multiscale object perception and suboptimal fine-grained boundary delineation. To address this, we specifically designed the progressive multiscale perception adaptor (PMSPA) and the spatial-frequency collaborative edge augmentation (SFCEA) module, enabling FastSAM to effectively meet the requirements of remote sensing-specific tasks and further improve the overall detection performance of the FastSAM-GAL. The main contributions of this study are summarized as follows.

1) A novel FastSAM-GAL framework is proposed for unsupervised multimodal CD. This framework maps multimodal imagery into a unified image domain through an adversarial learning strategy, effectively mitigating the incomparability problem induced by cross-modal heterogeneity.
2) The proposed FastSAM-GAL fully exploits the complementary information between general visual semantic features and remote sensing domain-specific features through a dual-branch architecture, thereby significantly enhancing the model's cross-scene generalization capability.
3) Two key functional modules, PMSPA and SFCEA, are designed to perform targeted remote sensing task adaptation and enhancement on the generic semantic features extracted by FastSAM, further improving the model's perceptual capability for land cover changes at varying granularities and its ability to delineate change boundaries.

The remainder of this article is organized as follows. Section II reviews related work on multimodal remote sensing CD and VFMs. Section III introduces the proposed FastSAM-GAL framework. Section IV presents and analyzes the results of comparative and ablation studies. Section V discusses the model's hyperparameter sensitivity, robustness, and computational efficiency. Section VI concludes this article.

II. Related Works

A. Multimodal CD

To address the incomparability issue of multimodal images, numerous methods have been proposed. Depending on whether neural networks are employed, these methods can be broadly categorized into traditional similarity-based approaches and deep-learning-based approaches. The following provides a brief review of the relevant research.

1) Traditional Similarity-Based Methods: These methods typically assume the existence of some intrinsic, imaging-independent relationship between multimodal images, and then use handcrafted functions to characterize these relationships, thereby constructing a comparable metric space. For example, Touati and Mignotte [26] defined similarity feature maps by assigning linear equation constraints to each pixel pair and efficiently solved them using multidimensional scaling techniques, ultimately integrating multiple automatic thresholding algorithms to generate CD results. Subsequently, Touati et al. [27] modeled the joint statistical distribution between different modalities based on pixel-pair observations and combined Markov random field spatial constraints with maximum a posteriori estimation to effectively extract change information. Sun et al. [28] posited that the same object shares identical structural information across different modal images, and based on this, they developed a multimodal CD method using an iterative robust graph. This method constructs a K-nearest neighbor graph to capture the internal structure between multimodal images and employs an iterative optimization strategy to progressively remove interference from changed regions, thereby refining cross-modal graph correspondences and improving CD accuracy. Inspired by this, Han et al. [29] proposed the adaptive optimization of the structured graph (AOSG) method, which deeply exploits comparable structural relationships between multimodal images to provide more reliable structural representations for difference comparison. Building on graph structures, Sun et al. [30] devised the similarity- and dissimilarity-induced image regression (SDIR) algorithm. This algorithm constructs both K-nearest neighbor and K-farthest neighbor graphs to regress the internal structure of one modality into the space of another modality, establishing cross-modal comparability, and subsequently applies a Markov segmentation model to perform multimodal CD. Similarly, Zheng et al. [31] developed a graffiti-guided structural regression fusion framework, which incorporates sparse graffiti annotations as priors into a dynamic regression mechanism, optimizing feature representations of key targets while suppressing irrelevant background information. Recently, Sun et al. combined superpixels with energy models to successively propose the locality-preserving energy model (LPEM) [32] and the rules-induced energy model (RIEM) [33], continuously deepening the exploitation and utilization of structural features in multimodal imagery from the perspective of energy optimization.

Overall, such methods are relatively simple to implement and offer strong interpretability. However, when dealing with large-scale image coverage or complex change scenarios, the design and implementation of specific metric functions often become considerably challenging.

2) Deep-Learning-Based Methods: These methods leverage the powerful nonlinear feature representation capabilities of neural networks to align multimodal images into a unified comparable feature domain or image domain, within

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which change information can be extracted. For example, Zhang et al. [34] reported a domain adaptation-based multimodal CD network, which aligns extracted deep features into a common feature space through a consistency constraint module, effectively alleviating cross-modal feature distribution discrepancies. The aligned features are then fed into a multiscale decoder to generate the final CD results. Subsequently, Hu et al. [35] developed a predict-refining-involved bidirectional contrastive difference network (PRBCD-Net). This network first introduces cross-reconstruction constraints to enhance the consistency of feature distributions across multimodal images. It then employs bidirectional contrastive constraints to reduce intraclass differences in unchanged regions while enlarging interclass differences in changed regions, thereby improving feature discriminability. To mitigate the adverse effects of image-domain noise on detection performance, Chen et al. [36] proposed a Fourier-domain structural relationship analysis framework, which effectively enhances the robust representation of cross-modal structural features by jointly modeling local and global structural relationships. Subsequently, they combined object-based image analysis techniques with an advanced Vision Transformer architecture and proposed a multimodal CD method named ObjFormer [37]. Furthermore, to further enhance the alignment effectiveness of cross-modal features, Liu et al. [38] proposed a hierarchical feature alignment network, embedding a multikernel maximum mean discrepancy-based alignment module at each level to more effectively regulate feature distributions across different layers. Liu et al. [39] argued that regardless of the image modality, unchanged regions consistently share more common features with one another compared with changed regions. Based on this viewpoint, they proposed the autoencoder Kolmogorov–Arnold network (AEKAN) [39], commonality feature representation learning (CFRL) [40], and dual-branch graph Kolmogorov–Arnold network (DGKAN) [41].

Different from the aforementioned approaches that perform change extraction by constructing a comparable feature space, Luppino et al. [42] approached the problem from the perspective of image translation and designed X-Net and ACE-Net, respectively. They mapped images acquired by different sensors into a unified image domain, such that the translated images can be treated as homogeneous image pairs, thereby enabling change extraction through direct comparison. Subsequently, Huang et al. [43] adopted NiceGAN [44] as the image translator and, on this basis, proposed a two-stage multimodal CD method. Li et al. [45] also utilized NiceGAN to mitigate the imaging discrepancies between optical and SAR images and further developed multiple attention mechanisms to enhance the model's ability to perceive change regions. Dong et al. [46] proposed a cycle translation-based collaborative training framework for multimodal CD, built upon the cross-modal image translation mechanism of CycleGAN. In addition, to reduce the interference of changed regions during the image translation process, Wu et al. [47] proposed a multidomain constrained translation network. This method introduces a contrastive learning mechanism to enforce consistency among neighboring pixels, thereby improving structure preservation capability. Inspired by this, Zhu et al. [48] integrated a dual-domain contrastive learning strategy into the image translation network to generate high-quality homogeneous image pairs, providing reliable inputs for subsequent CD. To thoroughly mine domain-invariant feature representations in multimodal images, Zhan et al. [49] designed a cross-domain difference representation learning (CDRL) framework. This framework consists of an image translation network and a CD network and achieves cross-modal alignment and difference modeling through joint optimization, thereby improving the accuracy and robustness of CD.

Although the aforementioned studies have achieved certain success in multimodal CD tasks, their feature representation capabilities remain highly dependent on specific data distributions due to the lack of universal prior knowledge. This limitation makes it difficult for these methods to maintain stable and reliable detection performance when dealing with multimodal remote sensing data across diverse change scenarios. In contrast, the proposed FastSAM-GAL effectively leverages the rich general prior knowledge embedded in VFMs, endowing it with stronger cross-modal feature representation capability and enhanced cross-scene generalization ability.

B. Vision Foundation Models

In recent years, VFMs represented by SAM [15], CLIP [50], and DINO [51] have demonstrated outstanding feature representation capabilities and strong cross-scene generalization potential in various downstream computer vision tasks, due to the rich general prior knowledge learned from large-scale pretraining data [52]. Among them, CLIP achieves performance comparable to fully supervised convolutional neural networks in zero-shot image classification tasks by leveraging contrastive learning on massive image–text pairs. This remarkable performance has attracted widespread attention from the research community, leading to the emergence of a series of VFM-based studies. For example, He et al. [53] proposed a visual masked training strategy based on the Vision Transformer [54], which achieves excellent self-supervised performance by randomly masking and reconstructing a large proportion of input image regions. Subsequently, Dong et al. [55] combined a masked self-distillation strategy with the CLIP framework, further enhancing the model's cross-modal representation learning capability.

Recently, researchers have begun to explore the incorporation of VFMs into unimodal remote sensing CD tasks, conducting preliminary investigations. For instance, Ding et al. [56] proposed a FastSAM-based CD framework, aiming to leverage the powerful visual representation capability of VFMs to improve the overall performance of CD in ultrahigh-resolution remote sensing images. Zheng et al. [57] introduced a segment anything for change model, which is capable of performing zero-shot prediction on unseen change types and unknown data distributions, demonstrating strong generalization potential. Subsequently, Zhang et al. [58] employed FastSAM as a bitemporal feature extraction encoder and further extended it with two dedicated modules to enhance the model's ability to perceive cross-scale fine boundaries and

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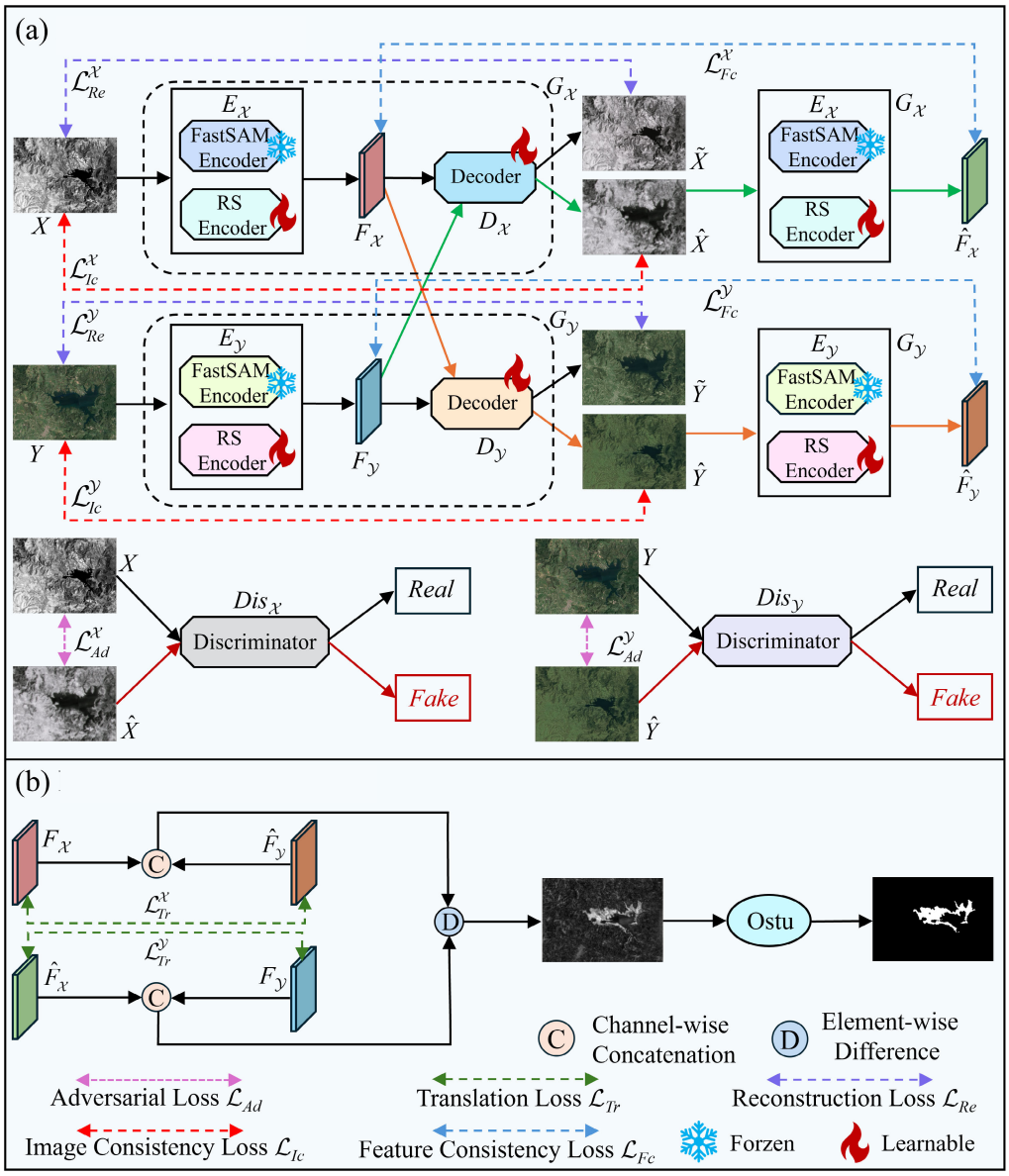


Fig. 2. Schematic of the overall FastSAM-GAL framework. (a) Detailed processing pipeline of adversarial learning. Multimodal images X and Y are taken as model inputs, and pseudo-target-domain images $\hat{Y}$ and $\hat{X}$ are generated through adversarial learning. Therein, X and $\hat{X}$ belong to the same image domain $\mathcal{X}$, while Y and $\hat{Y}$ belong to the same image domain $\mathcal{Y}$. (b) Binary change map generation process. Features within the same image domain are subjected to elementwise difference operations to produce a difference map, which is subsequently processed through Otsu thresholding to output a binary change result.

to recognize linear object targets. In addition, methods such as SAM-CR [59], SemiBCD [60], and DDSAM-CD [61] have also incorporated VFMs into CD networks through different architectural designs and learning strategies, further improving CD accuracy.

Although VFMs have demonstrated strong potential in unimodal CD tasks, research on their application in multimodal CD remains relatively limited, and their potential advantages have yet to be fully explored.

III. Methodology

A. Overview

In multimodal remote sensing CD research, two key challenges have consistently constrained its rapid development and widespread application. The first is how to effectively mitigate the inherent imaging discrepancies between images acquired from different sensors, ensuring good comparability between pre-event and postevent images. The second is how to learn feature representations with stronger generalization capability under limited data conditions, enabling the model to better adapt across different scenarios and maintain stable and superior detection performance on diverse datasets. To address the aforementioned challenges, this article proposes a novel FastSAM-GAL framework, which integrates the powerful representation capability of VFMs with the domain alignment mechanism of adversarial learning to achieve more accurate unsupervised multimodal CD. For convenience, the co-registered pre-event and postevent images are denoted as $X \in \mathbb{R}^{H\times W\times Cx}$ and $Y \in \mathbb{R}^{H\times W\times Cy}$, respectively, where H and W represent the height and width of the images, and $C_{\mathcal{X}}$ and $C_{\mathcal{Y}}$ denote the number of spectral bands of the corresponding images. Since X and Y are acquired by different sensors, they

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exhibit fundamental differences in imaging mechanisms and therefore belong to domain $\mathcal{X}$ and domain $\mathcal{Y}$, respectively.

As illustrated in Fig. 2, the proposed FastSAM-GAL framework primarily consists of two components. The first component introduces an adversarial learning strategy to map bitemporal images into each other's image domain, thereby effectively mitigating the incomparability problem caused by imaging discrepancies. Specifically, for the input bitemporal images $X \in \mathcal{X}$ and $Y \in \mathcal{Y}$, two structurally symmetric generators $G_{\mathcal{X}}$ and $G_{\mathcal{Y}}$ are constructed, respectively, to achieve cross-domain image translation. Correspondingly, discriminators $Dis_{\mathcal{X}}$ and $Dis_{\mathcal{Y}}$ are introduced to drive the generators through adversarial learning to produce more realistic domain translation results. Each generator is jointly composed of two parallel encoder branches and a decoder. The first encoder branch consists of a pretrained FastSAM model, whose weight parameters remain frozen throughout the entire training process. Benefiting from the rich visual priors acquired by FastSAM through large-scale pretraining, this branch is capable of extracting general semantic features with strong cross-scene generalization ability, such as spatial structural relationships and land-cover category semantics. These features can provide stable high-level semantic guidance for multimodal CD, thereby effectively alleviating the insufficient feature generalization caused by limited multimodal samples. The second branch employs a learnable remote sensing encoder, which focuses on modeling remote sensing-specific texture structures, spectral responses, and fine-grained land-cover differences, so as to enhance the model's domain adaptability to remote sensing scenarios. The two encoders form complementary collaboration at the feature level and jointly construct a dual-branch feature representation that balances generality and domain specificity.

Subsequently, the features from the two encoders are fused to obtain joint feature representations $F_{\mathcal{X}}$ and $F_{\mathcal{Y}}$, which balance semantic generalization and domain adaptation. The fused features are then fed into decoders $D_{\mathcal{X}}$ and $D_{\mathcal{Y}}$ to generate the cross-domain translated images $\hat{X}$ and $\hat{Y}$, as well as the within-domain reconstructed images $\tilde{X}$ and $\tilde{Y}$. Regarding the discriminators, since their design is not the primary focus of this study, the network structure from PatchGAN [62] is adopted directly. Based on the above network design, the cross-domain image translation process can be formally expressed as

$$\begin{aligned} \hat{X} &= D_{\mathcal{X}}\left(E_{\mathcal{Y}}(Y)\right), \quad \tilde{X} = D_{\mathcal{X}}\left(E_{\mathcal{X}}(X)\right) \cong X \\ \hat{Y} &= D_{\mathcal{Y}}\left(E_{\mathcal{X}}(X)\right), \quad \tilde{Y} = D_{\mathcal{Y}}\left(E_{\mathcal{Y}}(Y)\right) \cong Y \end{aligned} \tag{1}$$

where the pseudo-target-domain images $\hat{X} \in \mathcal{X}$ and $\hat{Y} \in \mathcal{Y}$ share the same visual style as the target images $X \in \mathcal{X}$ and $Y \in \mathcal{Y}$, and can be regarded as homogeneous image pairs.

After completing the first part of adversarial learning, the multimodal images have been converted into homogeneous image pairs, meaning that the bitemporal images now possess good comparability. Under this circumstance, the originally complex multimodal CD problem is effectively transformed into a conventional homogeneous CD task. At that moment, a common processing approach is to perform difference operations on bitemporal images within the same image domain (e.g., X and $\hat{X}$) to extract change information. However, this study does not adopt that strategy, because image-level differencing focuses only on low-level visual information, lacking the ability to perceive deeper semantic content of the objects, and thus struggles to distinguish real changes from appearance variations caused by imaging noise. Moreover, performing differencing directly at the image level would prevent the strongly generalized semantic features extracted by the FastSAM encoder from being fully utilized, thereby weakening the model's stability in complex scenarios.

In light of this, we choose to perform feature-level comparison in the aligned feature space, as a substitute for the conventional image-level differencing operation. Specifically, the original encoded features $F_{\mathcal{X}}$ of the pre-event image are concatenated with the aligned features $\hat{F}_{\mathcal{Y}}$ along the channel dimension, while the original encoded features $F_{\mathcal{Y}}$ of the postevent image are concatenated with the aligned features $\hat{F}_{\mathcal{X}}$ in the same manner, yielding dual-branch fused features. Elementwise differencing is then performed on these fused features to identify changes in land objects. Mathematically, this can be expressed as

$$DI = \sum_{i=1}^{C} \left(\left| Cat\left(F_{\mathcal{X}}, \hat{F}_{\mathcal{Y}}\right) - Cat\left(\hat{F}_{\mathcal{X}}, F_{\mathcal{Y}}\right) \right| \right) \tag{2}$$

where DI denotes the difference map, $Cat(\cdot)$ represents the channelwise concatenation operation, $|\cdot|$ denotes the absolute value operation, and C indicates the number of feature channels.

After obtaining the difference map DI, an adaptive thresholding method based on Otsu [63] is applied to binarize it, thereby generating the final CD result

$$CM(i,j) = \begin{cases} 1, & DI(i,j) > T \\ 0, & DI(i,j) \le T \end{cases} \tag{3}$$

where T denotes the optimal segmentation threshold.

B. Generator Architecture

The generator, serving as the core component of the proposed FastSAM-GAL framework, is primarily responsible for cross-domain mapping between multimodal images, and its architectural design directly influences the overall performance of CD. In this section, we take the image reconstruction process of the $\mathcal{Y}$ domain generator $G_{\mathcal{Y}}$ as an example to provide a detailed explanation of its network architecture and feature extraction mechanism. As depicted in Fig. 3, to prevent the general visual representations learned by the FastSAM encoder $E_{\mathcal{Y}}^{f}$ in $G_{\mathcal{Y}}$ from being disrupted during subsequent training, its weight parameters remain frozen throughout the entire training process. Moreover, compared with natural images, remote sensing images often contain multiple objects of varying scales, such as buildings, roads, and lakes. Since FastSAM is primarily designed for natural images, directly transferring it to remote sensing imagery makes it difficult to capture spatial information at different granularities. To address this, we designed the PMSPA module to further process the features $F_{\mathcal{Y}}^{f1}$, $F_{\mathcal{Y}}^{f2}$, $F_{\mathcal{Y}}^{f3}$, and $F_{\mathcal{Y}}^{f4}$ extracted from the FastSAM encoder, thereby effectively enhancing the model's ability to perceive

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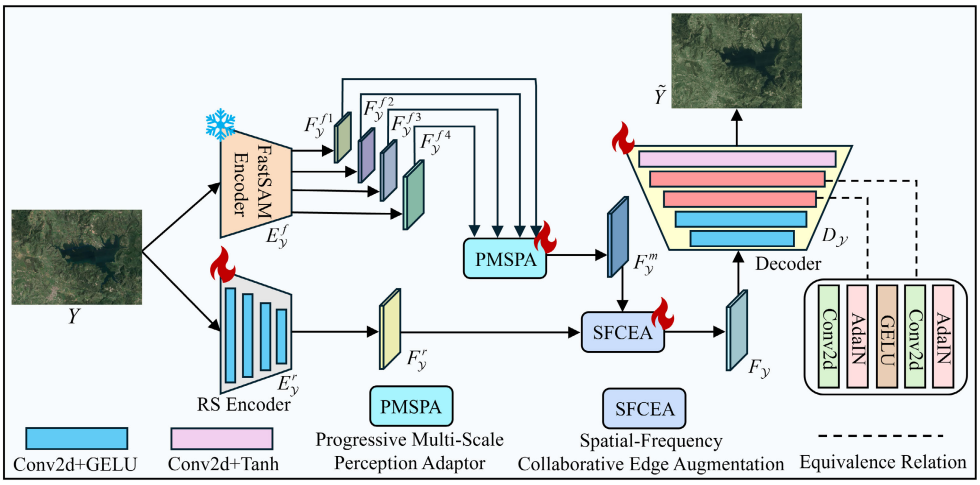


Fig. 3. Schematic of the generator architecture, primarily consisting of a FastSAM encoder with frozen parameters, a learnable remote sensing encoder, and a decoder.

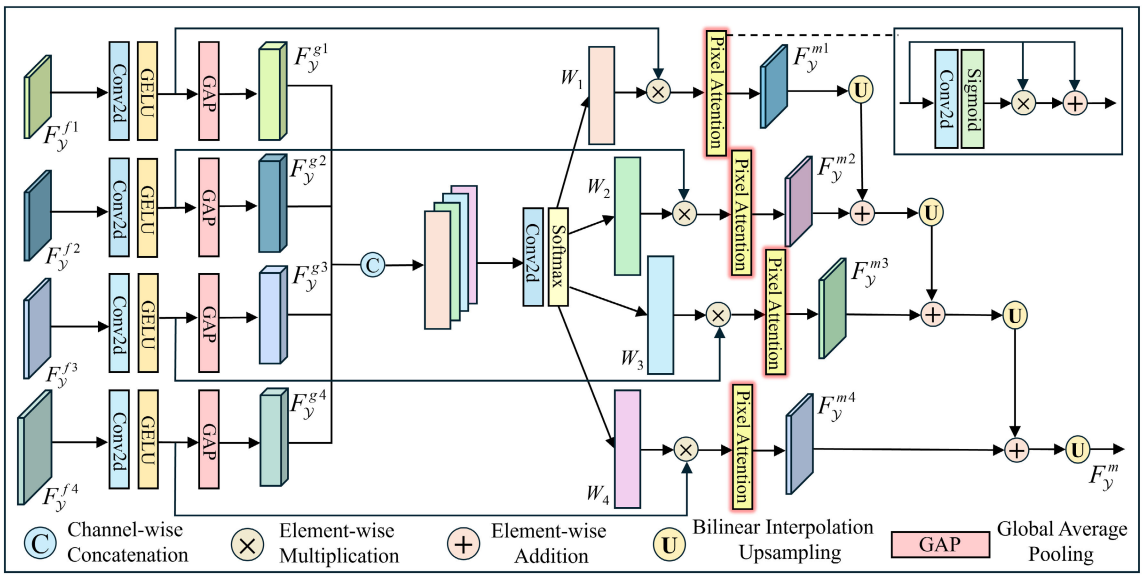


Fig. 4. Schematic of the PMSPA module structure.

spatial information at different granularities in remote sensing scenes. The implementation details of the PMSPA module will be described in Section III-C.

It is worth noting that, although the encoder $E_{\mathcal{Y}}^{f}$ extracts general semantic features with strong generalization ability, its pretraining is dominated by natural images. Therefore, it still has limitations in capturing patterns specific to the remote sensing domain, such as geometric distributions of objects and texture response characteristics. This inevitably constrains the model's cross-domain translation capability. To this end, we further introduce a learnable remote sensing encoder $E_{\mathcal{Y}}^{r}$. This encoder is constructed using 2-D convolution (Conv2d) layers and Gaussian error linear unit (GELU) activations as the basic building blocks, and it models remote sensing domain-specific features $F_{\mathcal{Y}}^{r}$ through layerwise nonlinear transformations, forming a complementary relationship with the general visual representations. After obtaining the outputs from the two encoders, we employ the SFCEA module to effectively fuse their complementary information and enhance the model's sensitivity to change boundaries. The detailed architecture of SFCEA is described in Section III-D. Finally, the fused features are fed into the decoder to reconstruct images in the corresponding domain. This process can be formalized as

$$\begin{cases} F_{\mathcal{Y}}^{r} = E_{\mathcal{Y}}^{r}(Y) \\ F_{\mathcal{Y}}^{f1}, F_{\mathcal{Y}}^{f2}, F_{\mathcal{Y}}^{f3}, F_{\mathcal{Y}}^{f4} = E_{\mathcal{Y}}^{f}(Y) \end{cases} \tag{4}$$

$$F_{\mathcal{Y}}^{m} = \mathcal{G}\left(F_{\mathcal{Y}}^{f1}, F_{\mathcal{Y}}^{f2}, F_{\mathcal{Y}}^{f3}, F_{\mathcal{Y}}^{f4}\right) \tag{5}$$

$$F_{\mathcal{Y}} = \mathcal{P}\left(F_{\mathcal{Y}}^{m}, F_{\mathcal{Y}}^{r}\right) \tag{6}$$

$$\tilde{Y} = D_{\mathcal{Y}}\left(F_{\mathcal{Y}}\right) \tag{7}$$

where $\mathcal{G}(\cdot)$ and $\mathcal{P}(\cdot)$ denote the feature mappings of the PMSPA and SFCEA modules, respectively. $F_{\mathcal{Y}}^{m}$ represents multiscale features containing information at different spatial granularities, while $F_{\mathcal{Y}}$ denotes the feature representation that integrates the general visual representations with the remote sensing domain-specific features.

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C. PMSPA Module

As shown in Fig. 4, the PMSPA module takes the four hierarchical features ($F_{\mathcal{Y}}^{f1} \in \mathbb{R}^{H/32\times W/32\times 640}$, $F_{\mathcal{Y}}^{f2} \in \mathbb{R}^{H/16\times W/16\times 640}$, $F_{\mathcal{Y}}^{f3} \in \mathbb{R}^{H/8\times W/8\times 320}$, and $F_{\mathcal{Y}}^{f4} \in \mathbb{R}^{H/16\times W/16\times 160}$) output by the encoder $E_{\mathcal{Y}}^{f}$ as input and effectively models object features of different granularities through a progressive multiscale aggregation mechanism. Specifically, the module first employs independent Conv2d layers and GELU activation functions to align the channel dimensions of features at each hierarchical level. The aligned features are then compressed into channelwise descriptors via global average pooling (GAP), capturing the global contextual information of each scale. Building on this, the module concatenates the four feature streams along the channel dimension and applies Conv2d followed by a Softmax function to dynamically generate the feature weights W_i for each level. These weights reflect the relative contributions of features at different scales, providing dynamic guidance for the adaptive calibration of multiscale features

$$W_1, W_2, W_3, W_4 = Softmax\left(Conv2d\left(Cat\left(F_{\mathcal{Y}}^{g1}, F_{\mathcal{Y}}^{g2}, F_{\mathcal{Y}}^{g3}, F_{\mathcal{Y}}^{g4}\right)\right)\right) \tag{8}$$

Subsequently, the generated weights are elementwise multiplied with the corresponding hierarchical features, guiding the model to focus on the granularity features most relevant to the current task. In addition, to suppress background noise and enhance the representation of key features, a pixel attention submodule is further introduced to refine the spatial responses of the calibrated features. Finally, the enhanced multiscale features are progressively upsampled, aligned at each stage, and sequentially summed to produce the aggregated feature $F_{\mathcal{Y}}^{m}$ with rich multigranularity semantic representation. It is worth emphasizing that, compared with directly upsampling to a uniform size, progressive upsampling effectively mitigates detail loss caused by scale discrepancies, thereby improving the overall quality of feature aggregation

$$F_{\mathcal{Y}}^{m} = \mathcal{U}\left(\mathcal{U}\left(\mathcal{U}\left(\mathcal{U}\left(F_{\mathcal{Y}}^{m1}\right) + F_{\mathcal{Y}}^{m2}\right) + F_{\mathcal{Y}}^{m3}\right) + F_{\mathcal{Y}}^{m4}\right) \tag{9}$$

where $F_{\mathcal{Y}}^{mi}$ denotes the multiscale features enhanced by the pixel attention module, with $i \in \{1, 2, 3, 4\}$ representing the feature hierarchy index. $\mathcal{U}(\cdot)$ represents the bilinear upsampling operation.

D. SFCEA Module

As previously described, the SFCEA module plays a dual role within the proposed FastSAM-GAL framework. On the one hand, SFCEA enhances the model's ability to recognize change boundaries by simultaneously capturing spatial edge information and high-frequency details in the frequency domain through the introduction of an edge detector and a 2-D fast Fourier transform (2D-FFT). As demonstrated in Fig. 5, the edge detector extracts structural contour priors from the original image, which are then refined by a learnable edge enhancement subnetwork to obtain more accurate edge representations. Meanwhile, the spatial-domain features are mapped to the frequency domain via Fourier transform, after which

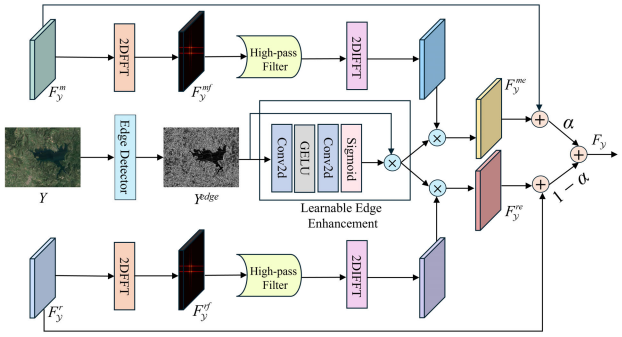


Fig. 5. Schematic of the SFCEA module structure. α is a learnable channelwise weight vector, which is constrained to the range of 0–1 through a Sigmoid mapping. During training, it adaptively adjusts the contribution ratio between general semantic features and remote sensing-specific domain features along the channel dimension, without requiring manual parameter tuning.

a high-pass filter removes low-frequency background redundancy, preserving high-frequency components rich in boundary and texture details. Finally, the high-frequency components are transformed back to the spatial domain and multiplied elementwise with the refined edge features, enabling the model to precisely focus on change boundaries in both spatial and frequency dimensions. This process can be expressed as

$$\begin{aligned} F_{\mathcal{Y}}^{me} &= \mathcal{F}^{-1}\left[\mathcal{F}\left(F_{\mathcal{Y}}^{m}\right)\cdot\mathcal{H}(u,v)\right]\cdot\mathcal{J}\left(Y^{edge}\right) \\ F_{\mathcal{Y}}^{re} &= \mathcal{F}^{-1}\left[\mathcal{F}\left(F_{\mathcal{Y}}^{r}\right)\cdot\mathcal{H}(u,v)\right]\cdot\mathcal{J}\left(Y^{edge}\right) \end{aligned} \tag{10}$$

where $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ denote the 2-D fast Fourier transform and its inverse, respectively; $\mathcal{H}(u, v)$ represents the high-pass filter function, where (u, v) are the frequency-domain coordinates. $\mathcal{J}(\cdot)$ denotes the learnable edge mapping, and $F_{\mathcal{Y}}^{me}$ and $F_{\mathcal{Y}}^{re}$ represent the general semantic features and the remote sensing features, respectively, after edge detail enhancement.

On the other hand, SFCEA is also responsible for the adaptive fusion of general semantic features and remote sensing-specific features. Since the general semantic features are derived from large-scale pretraining of the VFM, whereas the remote sensing-specific features are learned by an encoder tailored to remote sensing scenarios, they rely on different representation learning mechanisms. As a result, there exists a significant semantic gap between them in terms of feature distribution and semantic expression. If feature concatenation or simple addition is directly adopted, it is often difficult to effectively coordinate the complementary relationship between these two types of features. Moreover, such simple fusion strategies may directly superimpose inconsistent feature responses, thereby causing feature confusion, weakening feature representation capability, and ultimately degrading the final detection performance. To this end, SFCEA introduces a learnable channelwise weight vector $\alpha \in \mathbb{R}^{C}$, enabling the model to dynamically adjust the contributions of the two types of features according to the semantic importance of different channels. Specifically, when a certain channel relies more on the high-level semantic priors provided by the VFM, SFCEA enhances the response of the general semantic features. Conversely, when a channel depends more on local structures and fine-grained texture information in remote sensing scenarios, SFCEA increases the contribution of remote

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sensing-specific features. Through this dynamic weight allocation mechanism, SFCEA avoids the limitation of traditional fixed fusion strategies that assign uniform weights to all channels, thereby fully exploiting the complementary advantages of the two types of features in semantic representation and scene details. For the cth channel, the mathematical formulation is given as follows:

$$F_{\mathcal{Y}}^{(c)} = \alpha^{(c)} \cdot \left(F_{\mathcal{Y}}^{m,(c)} + F_{\mathcal{Y}}^{me,(c)}\right) + \left(1-\alpha^{(c)}\right) \cdot \left(F_{\mathcal{Y}}^{r,(c)} + F_{\mathcal{Y}}^{re,(c)}\right), \quad c = 1, \ldots, C \tag{11}$$

where $F_{\mathcal{Y}}^{c}$ represents the fused enhanced features of $F_{\mathcal{Y}}^{me,(c)}$ and $F_{\mathcal{Y}}^{re,(c)}$ at the cth channel. The value of $\alpha^{(c)}$ ranges from $[0, 1]$, with an initial value of 0.5.

E. Loss Function

To enhance the overall performance of FastSAM-GAL in multimodal CD, we design a series of loss functions to jointly optimize the model, including adversarial loss, reconstruction loss, image consistency loss, feature consistency loss, and translation loss. Their specific formulations are given as follows.

1) *Adversarial Loss $\mathcal{L}_{Ad}$:* By constructing an adversarial learning mechanism between the generator and the discriminator, the generator is driven to continuously improve, producing translated results that are highly consistent with the target-domain images in both visual style and statistical distribution

$$\mathcal{L}_{Ad}^{\mathcal{X}} = \mathbb{E}\left[Dis_{\mathcal{X}}(X)\right] + \mathbb{E}\left[1 - Dis_{\mathcal{X}}\left(\hat{X}\right)\right]$$
$$\mathcal{L}_{Ad}^{\mathcal{Y}} = \mathbb{E}\left[Dis_{\mathcal{Y}}(Y)\right] + \mathbb{E}\left[1 - Dis_{\mathcal{Y}}\left(\hat{Y}\right)\right] \tag{12}$$
$$\mathcal{L}_{Ad} = 0.5 \times \left(\mathcal{L}_{Ad}^{\mathcal{X}} + \mathcal{L}_{Ad}^{\mathcal{Y}}\right) \tag{13}$$

where $\mathbb{E}$ denotes the mathematical expectation.

2) *Reconstruction Loss $\mathcal{L}_{Re}$:* It is used to constrain the reconstructed images within the same domain to be consistent with the source-domain images at the pixel level, thereby preventing the generator from damaging the semantic content and spatial structure of the source images during cross-domain translation

$$\mathcal{L}_{Re} = 0.5 \times \left(\mathbb{E}\left[\left\|X - \tilde{X}\right\|_1\right] + \mathbb{E}\left[\left\|Y - \tilde{Y}\right\|_1\right]\right) \tag{14}$$

where $\|\cdot\|_1$ denotes the l_1-norm operation.

3) *Image Consistency Loss $\mathcal{L}_{Ic}$:* This primarily ensures that the translated images remain consistent with the target-domain images in unchanged regions. In other words, except for the areas with changes, the visual characteristics (such as texture patterns and structural properties) of all other regions should strictly correspond to those of the target-domain images

$$\mathcal{L}_{Ic} = 0.5 \times \left(\mathbb{E}\left[\left\|X - \hat{X}\right\|_2 \cdot P_u\right] + \mathbb{E}\left[\left\|Y - \hat{Y}\right\|_2 \cdot P_u\right]\right) \tag{15}$$

where $\|\cdot\|_2$ denotes the l_2-norm operation, and $P_u = 1\text{-}DI$ represents the probability of unchanged regions, where DI is computed according to (2).

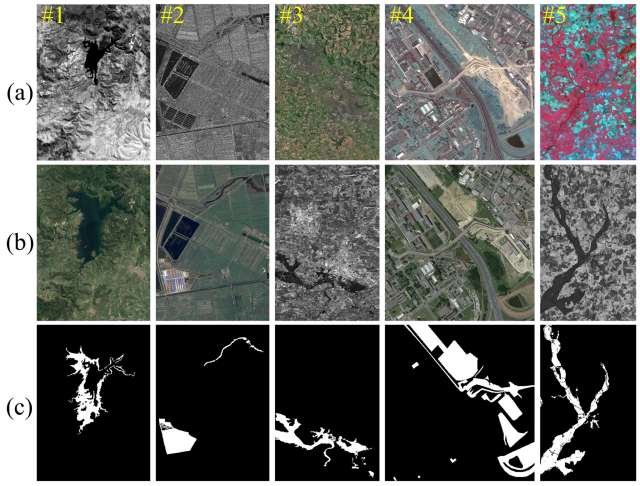


Fig. 6. Examples from datasets #1–#5. (a) Pre-event images. (b) Postevent images. (c) Ground-truth maps.

4) *Feature Consistency Loss $\mathcal{L}_{Fc}$:* By imposing a consistency constraint on the semantic features extracted by the encoder within the domain feature space, this loss ensures that the translated image achieves accurate semantic alignment with the target-domain image in the unchanged regions

$$\mathcal{L}_{Fc} = 0.5 \times \left(\mathbb{E}\left[\left\|F_{\mathcal{X}} - \hat{F}_{\mathcal{X}}\right\|_2 \cdot P_u\right] + \mathbb{E}\left[\left\|F_{\mathcal{Y}} - \hat{F}_{\mathcal{Y}}\right\|_2 \cdot P_u\right]\right) \tag{16}$$

5) *Translation Loss $\mathcal{L}_{Tr}$:* Although the same image may exhibit different visual styles across modalities, its underlying semantic content should remain consistent. Minimizing this loss enhances the accuracy of cross-domain mapping

$$\mathcal{L}_{Tr} = 0.5 \times \left(\mathbb{E}\left[\left\|F_{\mathcal{X}} - \hat{F}_{\mathcal{Y}}\right\|_2\right] + \mathbb{E}\left[\left\|F_{\mathcal{Y}} - \hat{F}_{\mathcal{X}}\right\|_2\right]\right) \tag{17}$$

In summary, the overall loss of FastSAM-GAL can be expressed as

$$\mathcal{L}_{Total} = \lambda\mathcal{L}_{Ad} + \mathcal{L}_{Re} + \mathcal{L}_{Ic} + \mathcal{L}_{Fc} + +\mathcal{L}_{Tr} \tag{18}$$

where λ denotes the weight coefficient for the adversarial loss, used to regulate its contribution in the joint optimization process. Based on extensive experiments, we set λ to 0.1 or 1, and its impact on CD performance will be discussed in detail in Section V-A.

IV. Experiments

A. Dataset Descriptions

As illustrated in Fig. 6, five widely used multimodal datasets [39] were employed to evaluate the effectiveness of the proposed FastSAM-GAL. The pre-event and postevent images in these datasets were acquired by different sensors, encompassing multiple imaging modalities such as optical, SAR, and multispectral data. Moreover, these datasets exhibit significant variations in image size, spatial resolution, geographic location, and types of changes. The change events involved include typical scenarios such as lake expansion, building

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TABLE I
DESCRIPTIONS OF THE EXPERIMENTAL DATASETS

Datasets	Date	Location	Size	Sensor	Event (Spatial Resolution)
#1	Sept. 1995 - July 1996	Sardinia, Italy	300×412×1(3)	Landsat5/ Google Earth	Lake Expansion (30 m)
#2	June 2008 - Sept. 2012	Shuguang, China	593×921×1(3)	Radarsat2/ Google Earth	Building Construction (8m)
#3	July 2006 - July 2007	Gloucester, England	2068×1164×3(1)	QuickBird2/ TerraSAR-X	Flooding (0.65m)
#4	May 2012 - July 2013	Toulouse, France	1000×1000×3(3)	Pleiades/ WorldView2	Construction (0.52m)
#5	1999 - 2000	Gloucester, England	990×554×3(1)	Spot/NDVI	Flooding (≈25m)

construction, and flood disasters, allowing for a comprehensive assessment of FastSAM-GAL's generalization ability across multiple dimensions. Detailed parameters for each dataset are listed in Table I.

B. Experimental Settings

1) Implementation Details: Given the relatively limited sample sizes of multimodal datasets, a sliding window strategy is employed during training to crop each dataset into 256×256 image patches. These patches were then subjected to data augmentation operations such as random rotation and random flipping to increase the diversity of training samples. For network optimization, the Adam optimizer ($\beta = 0.5$ and $\beta = 0.999$) was used to update the parameters of both the generators and discriminators, with learning rates set to 1×10^{-4} and 1×10^{-5}, respectively. The batch size was set to 2, and the number of training epochs was 20. During inference, the complete image was fed into FastSAM-GAL for change identification.

To evaluate the superiority of the proposed FastSAM-GAL, 12 state-of-the-art unsupervised multimodal CD approaches were selected as benchmarks, including DDCL-GAN [48], PRBCD-Net [35], SG-SRF [31], CGSL [64], SDIR [30], RIEM [33], LPEM [32], AOSG [29], MCFNet [65], AEKAN [39], CDRL [49], and CFRL [40]. To ensure fairness, all comparison methods strictly followed the model parameter settings provided in the original papers or their open-source implementations. The experiments were conducted on an Ubuntu 22.04 operating system. Deep-learning-based methods were implemented using the PyTorch framework, while traditional algorithms were carried out in MATLAB. The hardware platform consisted of an Intel Core i7-11700K processor (3.6 GHz), an NVIDIA GeForce RTX 3060 GPU (12-GB memory), and 64 GB of system RAM.

2) Evaluation Metrics: To quantitatively assess the performance of FastSAM-GAL, four metrics are adopted for comprehensive evaluation, including overall accuracy (OA), intersection over union (IoU), $F1$-score ($F1$), and the kappa coefficient (κ). All these metrics are computed based on the confusion matrix, consisting of true positives (TPs), false positives (FPs), true negatives (TNs), and false negatives (FNs). Their specific formulations are given as follows:

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \tag{19}$$

$$IoU = \frac{TP}{TP + FP + FN} \tag{20}$$

$$F1 = \frac{2TP}{2TP + FP + FN} \tag{21}$$

$$\kappa = \frac{OA - PRE}{1 - PRE} \tag{22}$$

$$PRE = \frac{(TP + FN)(TP + FP) + (TN + FP)(TN + FN)}{(TP + TN + FP + FN)^2} \tag{23}$$

where OA measures the overall classification accuracy of all pixels (both changed and unchanged). IoU and $F1$ focus on the detection accuracy of the changed class and are sensitive to both FPs and FNs, thereby providing a more objective reflection of the model's capability in identifying change regions. The κ corrects for random agreement by introducing PRE, enabling a more accurate reflection of the model's true detection performance under imbalanced data distributions.

C. Comparison Experiments

Fig. 7 presents the visualized CD results of different comparison methods on five multimodal datasets, with the corresponding quantitative evaluation results reported in Tables II and III. Overall, most comparison methods are significantly affected by modality discrepancies, resulting in a large number of FPs and FNs. This issue becomes particularly pronounced in complex change scenarios, where these methods often struggle to effectively distinguish real changes from background noise, leading to inaccurate insufficient capability in suppressing background noise, which leads to unchanged regions being incorrectly identified as changed areas. In contrast, the remaining methods show reduced false detections on this dataset, particularly PRBCD-Net and the proposed FastSAM-GAL, whose results are highly consistent with the actual changes. In Dataset #2, AOSG performs the worst, as its detection results miss a large portion of the primary changes and fail to effectively reflect the land-cover variations caused by building construction. This phenomenon can be attributed to the highly similar texture and structural characteristics of

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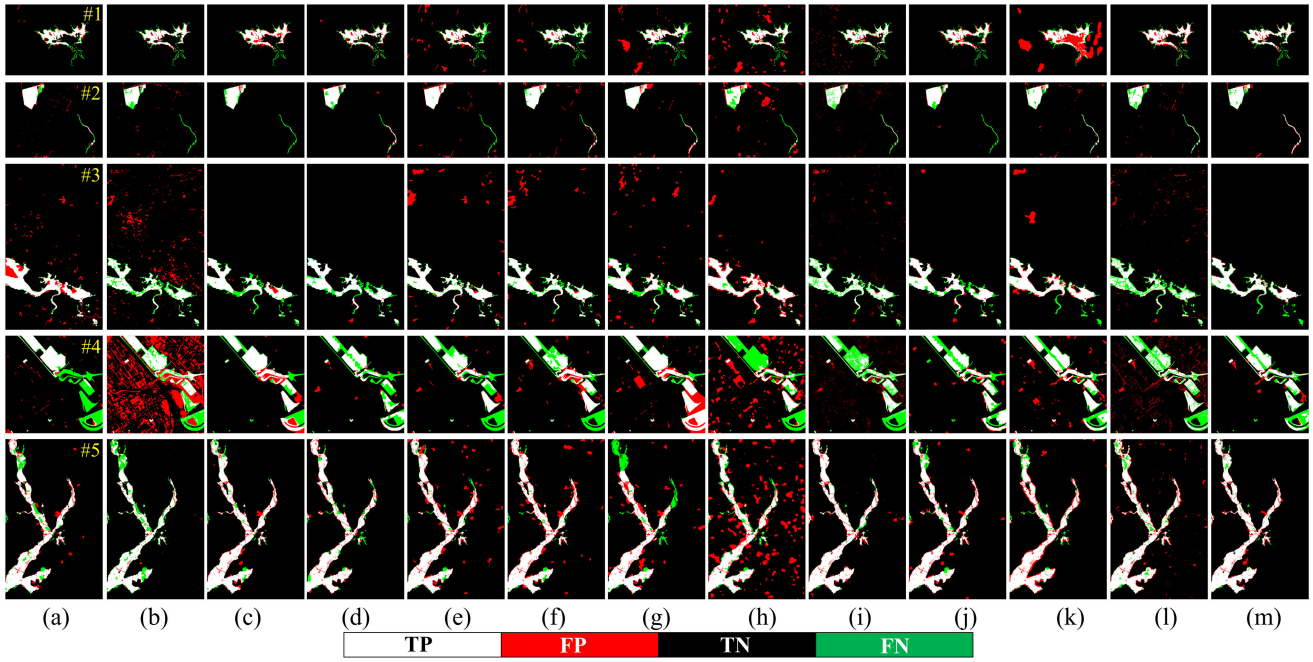


Fig. 7. Visual comparison of CD results obtained by different methods on five datasets (#1–#5). (a) DDCL-GAN [48]. (b) PRBCD-Net [35]. (c) SG-SRF [31]. (d) CGSL [64]. (e) SDIR [30]. (f) RIEM [33]. (g) LPEM [32]. (h) AOSG [29]. (i) MCFNet [65]. (j) AEKAN [39]. (k) CDRL [49]. (l) CFRL [40]. (m) FastSAM-GAL (Ours).

TABLE II

QUANTITATIVE COMPARISON OF CD RESULTS OF DIFFERENT METHODS ON DATASETS #1–#3 (BEST RESULTS ARE SHOWN IN BOLD, SECOND-BEST RESULTS ARE SHOWN IN UNDERLINED, AND ∗ DENOTES THE DEEP-LEARNING-BASED METHODS)

Methods	Dataset #1				Dataset #2				Dataset #3			
	OA	IoU	F1	κ	OA	IoU	F1	κ	OA	IoU	F1	κ
*DDCL-GAN	0.975	0.654	0.791	0.777	0.983	0.704	0.826	0.818	0.970	0.664	0.798	0.782
*PRBCD-Net	**0.978**	0.688	0.815	0.804	0.979	0.606	0.754	0.743	0.950	0.463	0.633	0.606
SG-SRF	0.972	0.642	0.782	0.767	**0.987**	0.731	0.845	0.838	0.980	0.702	0.825	0.814
*CGSL	0.976	0.667	0.800	0.787	0.986	0.732	0.845	0.838	**0.985**	0.748	0.856	0.848
SDIR	0.958	0.496	0.663	0.641	0.981	0.674	0.805	0.795	0.973	0.672	0.804	0.789
RIEM	0.971	0.594	0.745	0.730	0.979	0.654	0.791	0.780	0.973	0.657	0.793	0.779
LPEM	0.952	0.467	0.636	0.611	0.983	0.707	0.828	0.819	0.965	0.579	0.734	0.715
AOSG	0.942	0.442	0.613	0.582	0.951	0.377	0.548	0.522	0.968	0.649	0.787	0.770
*MCFNet	0.967	0.579	0.733	0.716	0.980	0.636	0.777	0.767	0.970	0.599	0.749	0.733
*AEKAN	0.968	0.591	0.743	0.726	**0.987**	0.721	0.838	0.831	0.983	0.735	0.847	0.839
*CDRL	0.888	0.304	0.467	0.416	0.984	0.706	0.828	0.819	0.972	0.655	0.791	0.776
*CFRL	0.973	0.662	0.797	0.783	0.977	0.575	0.730	0.718	0.962	0.497	0.664	0.644
*Ours	**0.978**	**0.691**	**0.817**	**0.806**	0.986	**0.745**	**0.854**	**0.846**	0.983	**0.754**	**0.860**	**0.851**

TABLE III

QUANTITATIVE COMPARISON OF CD RESULTS OF DIFFERENT METHODS ON DATASETS #4 AND #5 (BEST RESULTS ARE SHOWN IN BOLD, SECOND-BEST RESULTS ARE SHOWN IN UNDERLINED, AND ∗ DENOTES THE DEEP-LEARNING-BASED METHODS)

Methods	Dataset #4				Dataset #5				Average			
	OA	IoU	F1	κ	OA	IoU	F1	κ	OA	IoU	F1	κ
*DDCL-GAN	0.901	0.398	0.569	0.520	0.963	0.737	0.849	0.828	0.958	0.631	0.767	0.745
*PRBCD-Net	0.664	0.207	0.343	0.165	0.966	0.731	0.845	0.826	0.907	0.539	0.678	0.629
SG-SRF	0.922	0.582	0.736	0.690	0.971	0.788	0.881	0.865	0.966	0.689	0.814	0.795
*CGSL	0.921	0.515	0.680	0.637	0.971	0.785	0.880	0.863	0.968	0.689	0.812	0.795
SDIR	0.911	0.467	0.636	0.589	0.955	0.710	0.830	0.805	0.955	0.604	0.748	0.724
RIEM	0.903	0.463	0.633	0.579	0.958	0.725	0.841	0.817	0.957	0.619	0.761	0.737
LPEM	0.909	**0.589**	**0.741**	0.687	0.939	0.595	0.746	0.711	0.950	0.587	0.737	0.709
AOSG	0.804	0.191	0.321	0.207	0.894	0.488	0.656	0.597	0.912	0.429	0.585	0.536
*MCFNet	0.879	0.352	0.521	0.457	**0.975**	0.818	0.900	0.886	0.954	0.597	0.736	0.712
*AEKAN	0.924	0.555	0.714	0.671	0.964	0.750	0.857	0.837	0.965	0.670	0.800	0.781
*CDRL	0.914	0.545	0.706	0.656	0.955	0.702	0.825	0.800	0.943	0.582	0.723	0.693
*CFRL	0.885	0.412	0.583	0.518	0.960	0.731	0.845	0.822	0.951	0.575	0.724	0.697
*Ours	**0.929**	0.576	0.731	**0.691**	**0.975**	**0.822**	**0.903**	**0.888**	**0.970**	**0.718**	**0.833**	**0.816**

farmland regions in this scene, localization of the main change regions. In contrast, the proposed FastSAM-GAL demonstrates superior CD capability, producing clearer and more complete CD results across all datasets.

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Specifically, in Dataset #1, methods such as LPEM, AOSG, and CDRL exhibit noticeable FPs, indicating their which prevent the graph structure constructed by AOSG from capturing real changes. In addition, except for FastSAM-GAL, a changing river is almost entirely overlooked by all methods. This is likely due to the lack of effective mechanisms for modeling multigranularity changes, making it difficult for these methods to simultaneously perceive large-scale regional changes and fine-grained local variations. In contrast, FastSAM-GAL effectively overcomes this limitation by introducing the PMSPA module, demonstrating superior CD capability. For Datasets #3 and #5, the primary changes are caused by flood inundation. Due to the complex transitional regions between change boundaries and unchanged backgrounds, a large number of scattered FPs appear in the detection results of SDIR, RIEM, and LPEM. Furthermore, due to their limited ability to finely delineate change boundaries, the spatial integrity of the detected change regions is difficult to maintain.

In Dataset #4, under the complex urban construction scenario, PRBCD-Net produces large areas of FPs in unchanged regions, severely interfering with the identification of true change areas. In contrast, DDCL-GAN, AOSG, and MCFNet miss a substantial number of construction-related changes, resulting in evident FNs. This phenomenon reflects the limited capability of these methods to detect critical change information in complex scenes, making it difficult to effectively distinguish real changes from the dual interference of background noise and modality discrepancies. The underlying reason lies in the diverse and complex types of objects present in urban construction scenarios, where different land-cover categories exhibit significant variations in imaging characteristics across modalities. This makes it challenging for these methods to establish accurate cross-modal semantic correspondences, thereby hindering their ability to balance FP suppression and FN reduction. In contrast, the proposed FastSAM-GAL, benefiting from the rich semantic priors and cross-modal alignment capability provided by the VFM, is able to significantly reduce false detections while uncovering more true changes.

Tables II and III present the quantitative comparison results of the competing methods across different datasets. It can be observed that FastSAM-GAL achieves consistently superior and stable performance on all five datasets, without exhibiting the issue seen in PRBCD-Net, which performs well on certain datasets but experiences significant degradation on others. This demonstrates that the proposed method possesses strong generalization capability across different modality combinations and diverse scenarios. Notably, FastSAM-GAL attains average OA, IoU, $F1$, and κ values of 0.970, 0.718, 0.833, and 0.816 across the five datasets, respectively, significantly outperforming all other comparison methods and thereby fully validating its effectiveness and superiority in multimodal CD. Further analysis indicates that this performance advantage primarily stems from the synergistic optimization of cross-modal feature alignment and multiscale information modeling in FastSAM-GAL. On the one hand, by introducing a VFM to guide adversarial learning, the model can fully exploit highly generalizable semantic features, effectively alleviating the feature distribution shift caused by modality discrepancies. On the other hand, with the incorporation of PMSPA and SFCEA, the model is able to adaptively capture true change information at different spatial granularities, while jointly enhancing the fine delineation of change boundaries in both spatial and frequency domains, thereby effectively ensuring the spatial integrity and accuracy of the detected change regions.

D. Ablation Study

To further evaluate the contribution of key components in the proposed FastSAM-GAL to CD performance, we designed the following ablation experiments.

1) *Baseline:* It retains only the remote sensing encoder without introducing any additional modules.
2) *Only FastSAM:* It introduces the FastSAM encoder on top of the Baseline.
3) *FastSAM + PMSPA:* It incorporates both FastSAM and PMSPA based on the Baseline.
4) *FastSAM + SFCEA:* It incorporates both FastSAM and SFCEA based on the Baseline.
5) *FastSAM + Conv*1 $\times$ 1*:* Remote sensing features and generic semantic features are fused via channelwise concatenation followed by a 1×1 convolution, in order to compare the performance difference between the proposed SFCEA module and conventional feature fusion strategies.
6) *Full FastSAM-GAL:* The complete framework containing all components.

It should be noted that PMSPA and SFCEA are specifically designed to process the general features extracted by FastSAM, and therefore cannot function independently without FastSAM feature input. Consequently, ablation variants containing only PMSPA (Only PMSPA) or only SFCEA (Only SFCEA) were not designed. The effectiveness of these modules can be indirectly validated by comparing the experimental results of FastSAM + PMSPA and FastSAM + SFCEA.

We first evaluate the contribution of each major component to image translation performance. As illustrated in Fig. 8, the Baseline relies solely on remote sensing encoders for cross-modal feature modeling and does not benefit from the guidance of the generic features provided by FastSAM, such as land-cover semantic information and object-level structural representations. Consequently, the translated results exhibit noticeable deficiencies in both texture reconstruction and structural preservation. This limitation is particularly evident in Dataset #5, where the boundaries of flooded regions appear abrupt and unnatural, accompanied by blurred texture details. After introducing FastSAM, the model is able to leverage the generic semantic priors it provides to more accurately perceive object structures and regional distributions within the scene, thereby significantly improving the overall consistency of the translated images. It is worth noting that Only FastSAM, FastSAM + PMSPA, FastSAM + SFCEA, FastSAM + Conv1 $\times$ 1, and the full FastSAM-GAL employ the same backbone architecture and loss optimization strategy. As a result, some subtle differences among these variants are difficult to distinguish intuitively through visual inspection alone. Therefore, we further adopt the Fréchet inception distance

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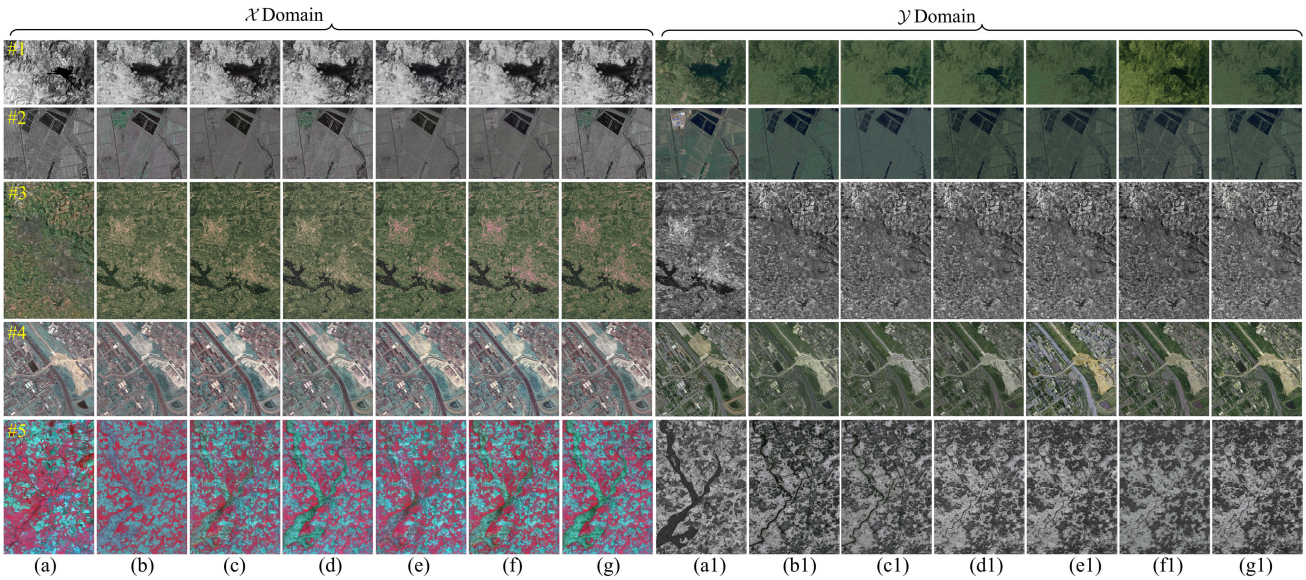


Fig. 8. Visualization of image translation results under different ablation settings. (a) and (a1) Original images. (b) and (b1) Baseline. (c) and (c1) Only FastSAM. (d) and (d1) FastSAM + PMSPA. (e) and (e1) FastSAM + SFCEA. (f) and (f1) FastSAM + Conv1 × 1. (g) and (g1) Full FastSAM-GAL.

TABLE IV

QUANTITATIVE IMAGE TRANSLATION RESULTS UNDER DIFFERENT ABLATION SETTINGS (BEST RESULTS ARE SHOWN IN BOLD, AND SECOND-BEST RESULTS ARE SHOWN IN UNDERLINED)

Conditions / FID	Dataset #1		Dataset #2		Dataset #3		Dataset #4		Dataset #5	
	$\mathcal{X}$	$\mathcal{Y}$	$\mathcal{X}$	$\mathcal{Y}$	$\mathcal{X}$	$\mathcal{Y}$	$\mathcal{X}$	$\mathcal{Y}$	$\mathcal{X}$	$\mathcal{Y}$
Baseline	264.13	281.39	233.06	354.90	291.50	381.23	282.10	281.98	175.00	278.45
Only FastSAM	254.18	276.32	276.36	385.63	267.75	370.23	272.35	271.67	150.28	277.91
FastSAM + PMSPA	<u>231.07</u>	<u>260.54</u>	<u>216.52</u>	<u>339.14</u>	<u>239.93</u>	<u>338.18</u>	<u>256.39</u>	<u>256.53</u>	<u>132.28</u>	<u>255.63</u>
FastSAM + SFCEA	237.44	262.09	254.39	367.45	254.39	349.98	261.34	265.81	141.26	256.64
FastSAM + Conv1×1	248.93	270.64	262.87	380.31	264.75	360.07	268.37	269.80	145.63	266.20
Full FastSAM-GAL	**223.51**	**255.84**	**210.41**	**333.39**	**222.97**	**305.02**	**248.92**	**239.08**	**128.54**	**251.72**

(FID) [66] to quantitatively evaluate the image translation performance of different variants, where a lower FID value indicates higher translation quality.

From the quantitative results shown in Table IV, incorporating PMSPA on top of FastSAM enables the model not only to effectively exploit generic semantic features but also to capture contextual relationships among objects at different scales, thereby achieving the second-best image translation performance. In particular, on Dataset #3, FastSAM + PMSPA attains FID scores of 239.93 and 338.18, representing a substantial improvement over the Baseline scores of 291.50 and 381.23, respectively. Furthermore, the full FastSAM-GAL, which integrates both PMSPA and SFCEA, achieves the best image translation performance across all datasets. This superiority can be attributed to the complementary strengths of PMSPA and SFCEA, which enhance the model's multiscale semantic representation capability and edge feature extraction capability, respectively. As a result, the model is able to more effectively reconstruct texture details at different granularities while preserving critical object structures.

Furthermore, we further evaluate the contribution of each component to the final CD performance. As demonstrated in Fig. 9(a)–(f), the Baseline relies solely on learning remote sensing domain-specific features, overlooking the semantic correlation information among different land-cover categories embedded in generic features, which results in scattered responses in the generated difference features. Although this approach can reflect the main change distribution to some extent, the contrast between changed and unchanged regions is low, limiting the model's ability to accurately distinguish true changes. When the FastSAM encoder is introduced (Only FastSAM), the response intensity in the change regions on the heatmap increases, indicating that incorporating general visual representations helps enhance the model's sensitivity to change targets. However, some regions still suffer from noticeable noise interference and blurred boundaries. Building on this, FastSAM + PMSPA further strengthens the representation of objects at different spatial scales through multiscale feature aggregation. The change regions on the heatmap become more concentrated and continuous, while background noise is effectively suppressed. FastSAM + SFCEA, on the other hand, enhances high-frequency details along change boundaries via joint spatial-frequency refinement, resulting in clearer and sharper boundary contours. Finally, the Full FastSAM-GAL, which integrates all components, produces the most concentrated and discriminative change responses on the heatmap, with optimal suppression of background noise.

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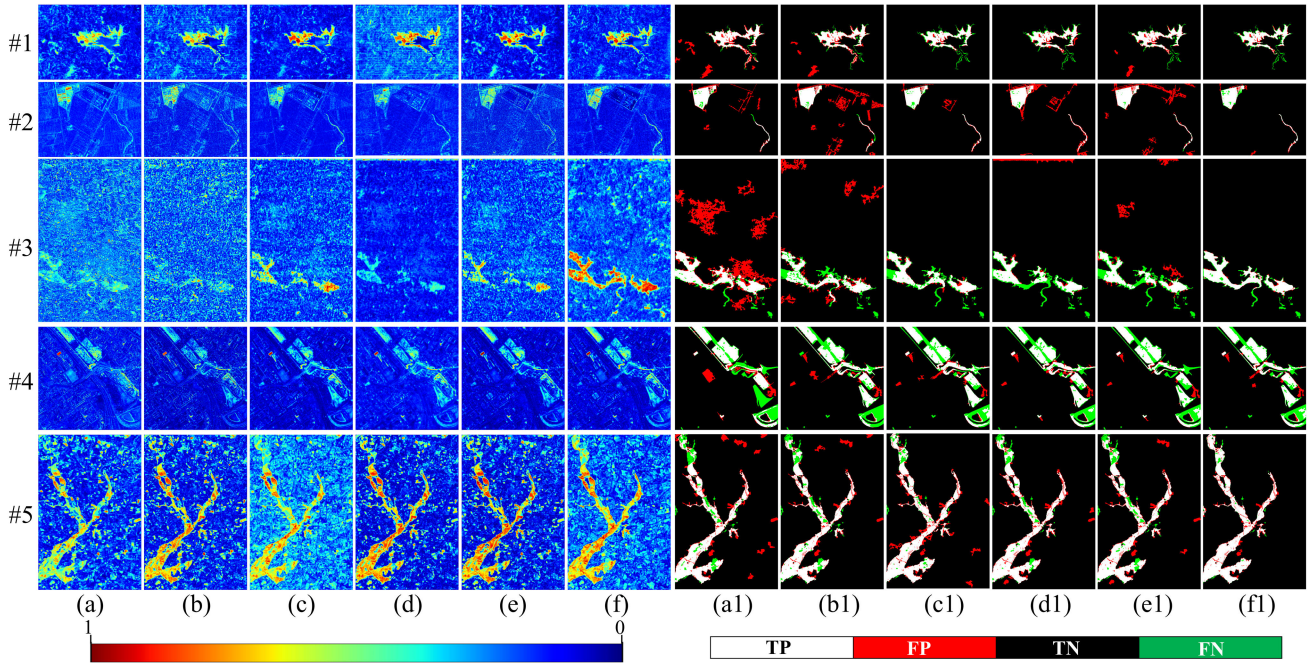


Fig. 9. Qualitative visualization of CD results under different ablation settings. (a) and (a1) Baseline. (b) and (b1) Only FastSAM. (c) and (c1) FastSAM + PMSPA. (d) and (d1) FastSAM + SFCEA. (e) and (e1) FastSAM + Conv1 × 1. (f) and (f1) Full FastSAM-GAL. (a)–(f) and (a1)–(f1) Difference feature heatmaps and the corresponding final CD results under different ablation settings, respectively.

TABLE V

QUANTITATIVE RESULTS OF DIFFERENT ABLATION SETTINGS (BEST RESULTS ARE SHOWN IN BOLD, AND SECOND-BEST RESULTS ARE SHOWN IN UNDERLINED)

Conditions	Dataset #1		Dataset #2		Dataset #3		Dataset #4		Dataset #5	
	F1	κ	F1	κ	F1	κ	F1	κ	F1	κ
Baseline	0.718	0.696	0.754	0.741	0.480	0.428	0.630	0.576	0.766	0.733
Only FastSAM	0.757	0.738	0.541	0.509	0.576	0.544	0.677	0.630	0.806	0.779
FastSAM + PMSPA	<u>0.798</u>	<u>0.786</u>	<u>0.826</u>	<u>0.817</u>	<u>0.757</u>	<u>0.743</u>	<u>0.719</u>	<u>0.677</u>	<u>0.829</u>	<u>0.803</u>
FastSAM + SFCEA	0.788	0.775	0.654	0.632	0.669	0.648	0.715	0.673	0.819	0.793
FastSAM + Conv1×1	0.783	0.768	0.651	0.628	0.598	0.573	0.699	0.658	0.809	0.782
Full FastSAM-GAL	**0.817**	**0.806**	**0.854**	**0.846**	**0.860**	**0.851**	**0.731**	**0.691**	**0.903**	**0.888**

From the final CD results shown in Fig. 9(a1)–(f1), it can be observed that with the introduction of each component, the areas of FPs and FNs exhibit a significant decreasing trend. However, this trend is not reflected in Dataset #2, where the Baseline's detection results are noticeably better than those of Only FastSAM and FastSAM + SFCEA. This anomaly primarily stems from the abundance of farmland in this dataset, whose visual representations differ substantially between SAR and optical imagery. Specifically, SAR imagery primarily captures the backscattering characteristics of land-cover objects, which are susceptible to surface roughness, geometric structure, and speckle noise; whereas optical imagery mainly reflects the spectral reflectance properties of surface objects, conveying more intuitive semantic information in terms of color, texture, and appearance. Consequently, the same farmland object tends to exhibit significant discrepancies in texture patterns, edge structures, and semantic representations between SAR and optical imagery, resulting in pronounced cross-modal feature distribution shifts. Moreover, the FastSAM encoder remains frozen throughout training, and its output generic visual features are predominantly derived from large-scale natural image pretraining. Such feature representations are more biased toward color, texture, and edge priors in optical scenes, making it difficult to effectively adapt to the specific semantic distributions of diverse remote sensing scenarios without further learning and adjustment via PMSPA. In this case, directly introducing general visual features and remote sensing domain-specific features into the CD network not only fails to fully exploit their complementary advantages but may also negatively affect detection performance due to semantic conflicts between heterogeneous feature representations. In contrast, after incorporating PMSPA (i.e., FastSAM + PMSPA), the proposed PMSPA performs remote sensing-aware semantic adaptation and feature alignment on the frozen FastSAM features, enabling the general visual priors to gradually adapt to the target remote sensing features. This process effectively alleviates the semantic conflicts caused by cross-modal distribution discrepancies. As a result, the model is able to preserve the high-level semantic knowledge provided by the VFM while simultaneously

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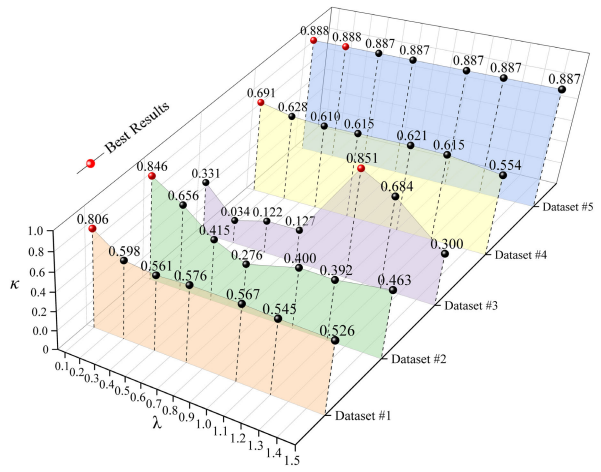


Fig. 10. Effect of different λ settings on CD performance.

enhancing its compatibility and consistency with remote sensing scene characteristics. Consequently, the detection performance is significantly improved after the introduction of PMSPA, which strongly validates the necessity and effectiveness of the proposed PMSPA design.

Furthermore, although both SFCEA and Conv1 $\times$ 1 can achieve the fusion of remote sensing features and general semantic features, their fusion effectiveness and overall performance differ due to the distinct feature fusion mechanisms. Specifically, SFCEA introduces learnable channelwise weight vectors, enabling the model to dynamically adjust the contribution of different features according to their semantic importance across channels, thereby fully exploiting the complementary information among heterogeneous features. In contrast, Conv1 $\times$ 1 achieves feature fusion merely through simple channel concatenation and linear mapping, which struggles to effectively establish deep correlations between features from different modalities. Consequently, FastSAM + SFCEA outperforms FastSAM + Conv1 $\times$ 1 across the FID, $F1$, and κ metrics, confirming the superiority of SFCEA in cross-modal feature fusion. It is worth noting that when FastSAM, PMSPA, and SFCEA are all incorporated (i.e., Full FastSAM-GAL), the detection results align even more closely with the true changes. As shown in the quantitative results in Table V, FastSAM + PMSPA and Full FastSAM-GAL achieve the second-best and best performance, respectively. This clearly indicates that each component contributes positively to enhancing CD performance, and their collaborative optimization can further strengthen the overall capability of the model.

V. Discussion

A. Hyperparameter Analysis

This section conducts a sensitivity analysis of the hyperparameter λ in (18). This parameter primarily influences the model's performance in the CD task by adjusting the weight of the adversarial loss within the overall optimization objective. Specifically, λ is set to 0.1, 0.3, 0.5, 0.7, 1.0, 1.2, and 1.5, and the model's performance is systematically evaluated under the same experimental settings to determine the optimal hyperparameter configuration.

As depicted in Fig. 10, on Datasets #1, #2, #4, and #5, the model achieves optimal detection performance when $\lambda = 0.1$. As λ gradually increases, performance on these datasets does not improve further. This is likely because when the adversarial loss dominates the total loss, the optimization of other loss terms is suppressed, which in turn has a certain negative impact on CD performance. In contrast to the patterns observed above, the performance trend on Dataset #3 shows a distinct difference. Its κ metric performs poorly when λ is small, whereas the model achieves the best performance at $\lambda = 1.0$. This phenomenon may be closely related to the larger scene coverage and higher spatial resolution of Dataset #3. A larger coverage area usually implies more diverse land-cover types, more complex spatial structures, and more pronounced cross-regional distribution differences within the scene. In this case, a relatively weak adversarial constraint may be insufficient to effectively drive the generator to perform cross-modal domain mapping in complex scenarios, thereby limiting the reliability of subsequent CD results.

Overall, the optimal value of λ is closely related to the scene complexity, coverage area, and spatial resolution of the dataset. For datasets with moderate coverage and relatively low scene complexity, such as Datasets #1, #2, #4, and #5, λ is recommended to be set to 0.1 to avoid excessive adversarial constraints interfering with the joint optimization of other loss terms. In contrast, for scenarios with characteristics similar to Dataset #3, namely, datasets with larger coverage areas and more complex land-cover types, λ is recommended to be set to 1.0 to fully activate the optimization potential of adversarial training, thereby improving the quality of cross-modal mapping and the CD performance in complex scenarios.

B. Features Visual Analysis

Cross-domain image translation is an important approach for addressing the incomparability of multimodal images [67]. In this section, the feature distributions of the translated images are visualized using t-SNE, and the degree of feature alignment is quantitatively evaluated using the relative root mean square (RRMS) metric to validate the effectiveness of FastSAM-GAL in mitigating imaging differences. As shown in Fig. 11, the generated translated images ($\hat{X}$ and $\hat{Y}$) exhibit high consistency with the real target-domain images (X and Y) in terms of overall visual style, land-cover texture, and spatial structure, indicating that the proposed FastSAM-GAL can effectively reduce the distribution gap between different modalities.

From the t-SNE feature distribution visualization within the white rectangular regions, the original multimodal image features [see Fig. 11(d)] exhibit clear cluster separation, reflecting the inherent modality differences and feature misalignment between images captured by different sensors. After cross-domain translation via FastSAM-GAL, the features of the translated images show a higher degree of overlap with the target-domain features in the embedding space [see Fig. 11(e) and (f)], demonstrating effective cross-modal alignment. Meanwhile, the quantitative RRMS results further support this observation. The RRMS values for all datasets decrease to varying degrees after translation, with Dataset #1 showing the most significant improvement: the RRMS between the

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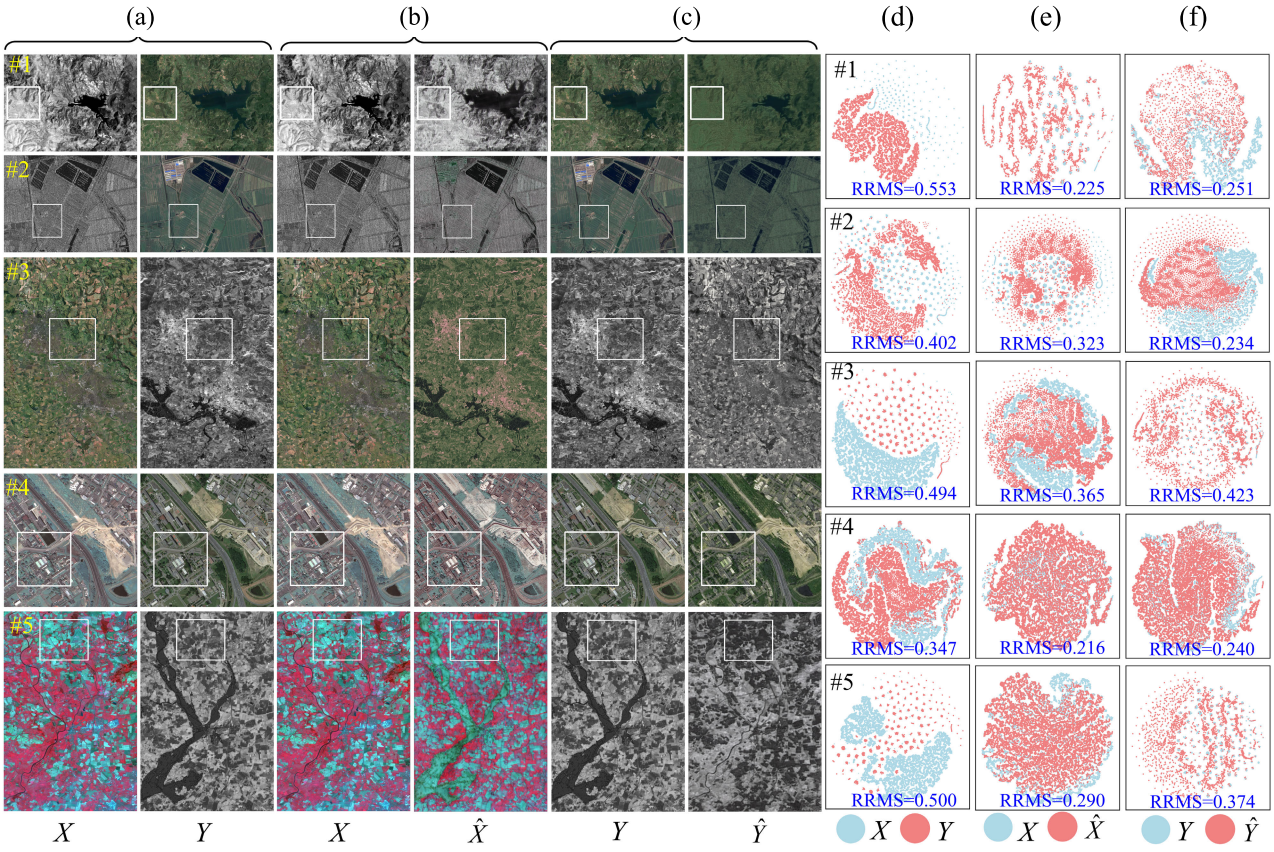


Fig. 11. Cross-domain image translation results and t-SNE visualization of feature distributions for the unchanged regions within the white rectangular boxes. (a) Original multimodal image pairs. (b) Translated image pairs in domain $\mathcal{X}$. (c) Translated image pairs in domain $\mathcal{Y}$. (d) Feature distribution of original multimodal images. (e) Feature distribution of translated images in domain $\mathcal{X}$. (f) Feature distribution of translated images in domain $\mathcal{Y}$.

$\mathcal{X}$ domain and $\mathcal{Y}$ domain drops from 0.553 in the original images to 0.225 and 0.251, respectively. Together, these qualitative and quantitative results confirm the effectiveness of the proposed FastSAM-GAL in mitigating multimodal image distribution differences and achieving accurate feature alignment.

C. Generalization Ability Analysis

To further validate the generalization ability of the proposed FastSAM-GAL, systematic experiments were conducted on multimodal images selected from four different regions of the BRIGHT [68] dataset, including Beta, Congo, Hawaii, and Türkiye. In these regions, the pre-event images were acquired by optical sensors, while the postevent images were collected by SAR sensors, forming representative multimodal CD scenarios.

As shown in Fig. 12, although different regions exhibit significant variations in land-cover distribution and scene structure, the proposed FastSAM-GAL can still accurately identify the main changed regions, with noticeably fewer detection errors than the comparison methods. Specifically, in the Congo region, the changed targets are scattered, small in size, and located in complex backgrounds, which imposes high requirements on the fine-grained discrimination capability of CD methods. Some comparison methods, such as MCFNet and AEKAN, suffer from considerable missed detections, whereas FastSAM-GAL can preserve fine-grained changed targets more completely while maintaining a low false alarm rate.

In addition, since these comparison methods lack the general semantic features provided by VFMs, they exhibit certain limitations in cross-scene generalization. Specifically, although CGSL achieves relatively good detection performance in the Beta region, it produces a large number of FPs and FNs in the Hawaii region. This indicates that CGSL is sensitive to changes in scene distribution, and its feature-representation capability may degrade to some extent when land-cover types and modal discrepancies vary significantly. A similar phenomenon can also be observed for CDRL, which can effectively locate the main changed regions in the Congo region but suffers from a significant performance decline in Türkiye region. These observations suggest that when change modeling relies solely on remote sensing-specific features, the model may struggle to fully learn high-level semantic representations shared across different regions, making it vulnerable to scene distribution shifts in cross-regional scenarios.

It is worth noting that the proposed FastSAM-GAL introduces general semantic priors provided by the VFM, enabling the model to simultaneously exploit remote sensing-specific detailed information and shared semantic representations across scenes. As a result, FastSAM-GAL maintains more stable and reliable detection performance across different regions and complex multimodal scenarios. From the quantitative results in Table VI, the proposed method achieves

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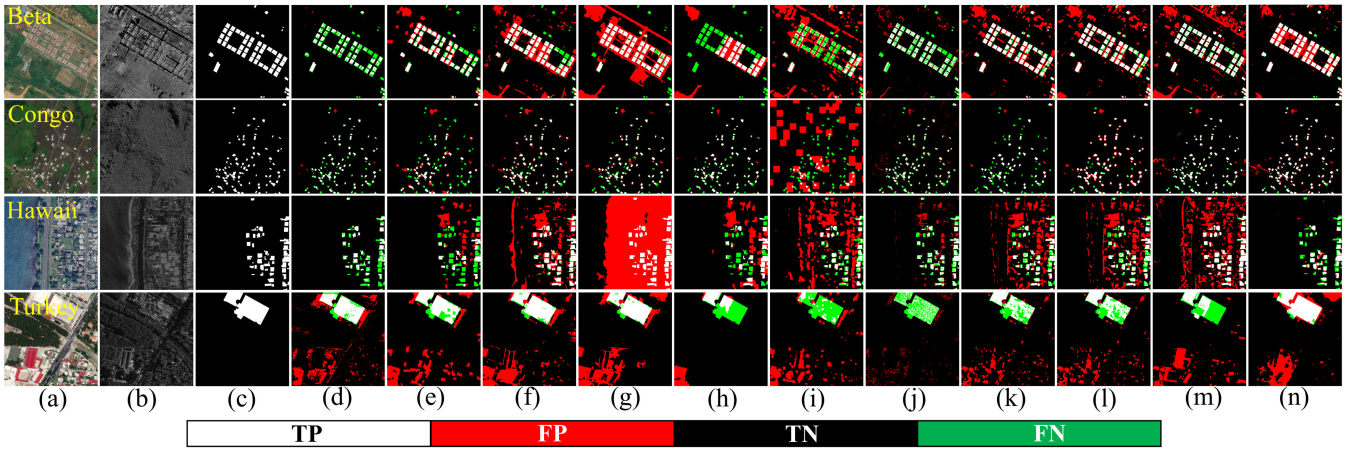


Fig. 12. Qualitative comparison of CD results across different regions. (a) Pre-event image. (b) Postevent image. (c) Ground truth. (d) DDCL-GAN [48]. (e) CGSL [64]. (f) SDIR [30]. (g) RIEM [33]. (h) LPEM [32]. (i) AOSG [29]. (j) MCFNet [65]. (k) AEKAN [39]. (l) CDRL [49]. (m) CFRL [40]. (n) FastSAM-GAL (Ours).

TABLE VI

QUANTITATIVE EVALUATION OF CD RESULTS ACROSS DIFFERENT SCENARIOS (BEST RESULTS ARE SHOWN IN BOLD, SECOND-BEST RESULTS ARE SHOWN IN UNDERLINED, AND ∗ DENOTES THE DEEP-LEARNING-BASED METHODS)

Methods	Beta			Congo			Hawaii			Turkey			Average		
	OA	F1	κ	OA	F1	κ	OA	F1	κ	OA	F1	κ	OA	F1	κ
*DDCL-GAN	**0.947**	0.634	0.609	0.980	0.720	0.711	**0.963**	0.631	0.614	0.878	0.522	0.462	**0.942**	0.627	0.599
*CGSL	0.929	0.668	0.628	0.963	0.527	0.508	0.913	0.339	0.292	0.882	0.518	0.458	0.922	0.513	0.472
SDIR	0.862	0.531	0.461	0.974	0.694	0.680	0.871	0.478	0.421	0.872	0.535	0.473	0.895	0.560	0.509
RIEM	0.768	0.416	0.315	0.973	0.671	0.364	0.364	0.174	0.059	0.855	0.510	0.443	0.740	0.443	0.295
LPEM	0.877	0.423	0.355	**0.981**	0.727	0.717	0.888	0.375	0.318	0.919	0.435	0.392	0.916	0.490	0.446
AOSG	0.781	0.263	0.152	0.778	0.090	0.022	0.815	0.287	0.208	0.838	0.148	0.060	0.803	0.197	0.111
*MCFNet	0.934	0.623	0.587	0.968	0.606	0.589	0.944	0.598	0.568	**0.923**	0.432	0.393	0.942	0.565	0.534
*AEKAN	0.916	0.560	0.513	0.977	0.644	0.633	0.915	0.105	0.068	0.920	0.459	0.417	0.932	0.442	0.408
*CFRL	0.884	0.592	0.532	0.970	0.689	0.674	0.864	0.381	0.318	0.910	0.540	0.491	0.907	0.551	0.504
*CDRL	0.859	0.544	0.474	0.977	0.730	0.718	0.826	0.416	0.349	0.876	0.346	0.279	0.885	0.509	0.455
*Ours	0.918	**0.688**	**0.645**	0.976	**0.747**	**0.735**	0.961	**0.642**	**0.622**	0.884	**0.579**	**0.523**	0.935	**0.664**	**0.632**

the best performance in terms of the key evaluation metrics, including $F1$ and κ, on all datasets. This further demonstrates the strong generalization capability of the proposed method for multimodal CD tasks across different regions.

D. Robustness Analysis

Multimodal CD, not constrained by the requirement that bitemporal images come from the same sensor, has been widely applied to rapid emergency response for sudden events such as flood inundation, forest fires, and earthquakes. However, such events are often accompanied by extreme weather conditions. On the one hand, this can interfere with the echo signals of SAR sensors, exacerbating speckle noise in SAR images; on the other hand, it may induce significant spectral distortions, leading to the degradation of land-cover features in optical images. These adverse imaging conditions pose serious challenges to the robustness of CD models in real-world emergency response scenarios.

To evaluate the interference resistance and robustness of the proposed FastSAM-GAL under complex weather conditions, this section follows the setup in [49] and simulates various optical image degradation scenarios (cloudy, foggy, and strong illumination) and SAR image noise scenarios (low, medium, and high speckle noise) on Dataset #3. The corresponding experimental results are shown in Fig. 13. Under cloudy and foggy conditions [see Fig. 13(a) and (b)], the texture details of land-cover features in the optical images are partially obscured or blurred, and the overall contrast is significantly reduced. Nevertheless, FastSAM-GAL is still able to accurately identify the main change regions, with only minor FPs and FNs appearing along the change boundaries. Under strong illumination conditions [see Fig. 13(c)], the saturation and brightness of the optical images increase significantly, causing substantial shifts in the spectral characteristics of land-cover features. FastSAM-GAL still demonstrates strong interference resistance, maintaining stable CD performance even under severe spectral distortions. Regarding the SAR image noise scenarios, as the speckle noise intensity gradually increases [see Fig. 13(d)–(f)], the structural information of land-cover features in the SAR images becomes increasingly obscured, and the overall image quality deteriorates. Nevertheless, under these conditions, the CD results produced by FastSAM-GAL still preserve spatial patterns that closely match the actual changes, demonstrating excellent noise robustness.

The quantitative results in Table VII further validate the above observations. Under six adverse imaging conditions, the average $F1$ and κ reach 0.834 and 0.826, respectively, representing only a 2.3% and 2.4% decrease compared with

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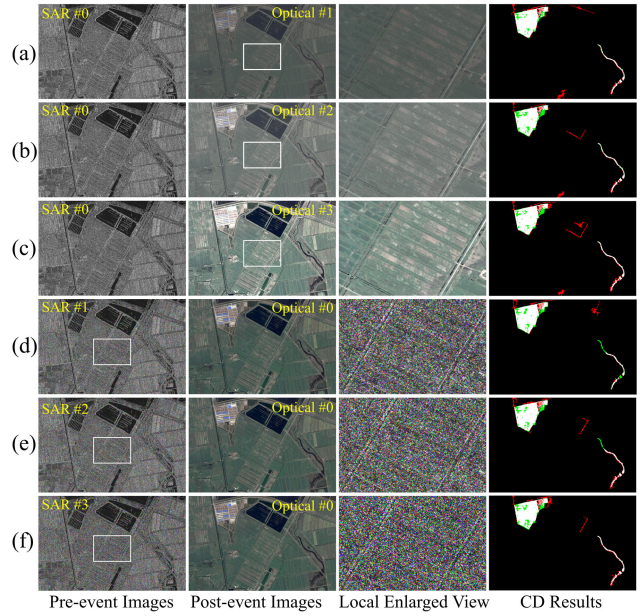


Fig. 13. Simulated optical and SAR images under various adverse conditions on Dataset #3. (a) Optical #1 cloudy scenario ($\sigma = 0.8$, $\varphi = 0.7$, and $\omega = 0.8$). (b) Optical #2, foggy scenario ($\sigma = 1.3$, $\varphi = 0.7$, and $\omega = 0.5$). (c) Optical #3, daytime scenario ($\sigma = 1.5$, $\varphi = 1.2$, and $\omega = 0.6$), where σ, φ, and ω denote the brightness scaling factor, contrast coefficient, and saturation adjustment factor, respectively. (d) SAR#1, image with low speckle noise ($\gamma = 6$). (e) SAR#2, image with moderate speckle noise ($\gamma = 4$). (f) SAR#3, image with severe speckle noise ($\gamma = 2$), where γ represents the shape parameter controlling speckle noise simulation. Optical#0 and SAR#0 correspond to the original images.

TABLE VII
QUANTITATIVE COMPARISON UNDER ADVERSE CONDITIONS ON DATASET #2

Pre-event Image	Post-event Image	F1	κ
Original SAR#0	Original Optical#0	0.854	0.846
Original SAR#0	Optical#1	0.842	0.834
	Optical#2	0.831	0.823
	Optical#3	0.821	0.812
SAR#1	Original Optical#0	0.842	0.834
SAR#2		0.837	0.829
SAR#3		0.834	0.826
Average		0.834(−2.3%)	0.826(−2.4%)

the original imaging conditions. This demonstrates that the proposed FastSAM-GAL can maintain highly stable CD performance even under challenging interference conditions. This robustness advantage is primarily attributed to the strong generalization capability of the VFM, which enables the network to preserve stable high-level semantic representations even when image quality deteriorates, effectively mitigating the impact of adverse conditions on feature extraction.

E. Time Efficiency Analysis

To comprehensively assess the practical applicability of the proposed FastSAM-GAL framework, this section further analyzes the inference efficiency of all compared methods in addition to detection accuracy. Table VIII reports the inference time and corresponding performance of each method on Dataset #2 ($593 \times 921 \times 1(3)$) under the same hardware environment. There is a considerable variation in time costs among the methods. AOSG, due to its complex adaptive graph structure construction, requires up to 34.3 s, making it the most time-consuming method. SDIR takes 28.6 s, with its efficiency bottleneck mainly caused by superpixel segmentation and bidirectional regression computations. MCFNet requires 27.1 s, also facing a significant efficiency challenge. In contrast, AEKAN and CDRL achieve relatively high time efficiency, with inference times of 1.6 and 1.5 s, respectively. However, their κ are 0.831 and 0.819, respectively, which still exhibit a noticeable performance gap compared with FastSAM-GAL.

Furthermore, CFRL adopts a lightweight network architecture that reduces the model parameter count to 0.2 M while achieving an optimal inference speed of 0.2 s. It is worth noting that the design objective of the proposed FastSAM-GAL is not lightweight modeling, but rather focuses on enhancing the feature representation capability and cross-scene generalization performance of the model under complex scenarios. Consequently, the incorporation of VFM and associated adaptation modules inevitably increases model complexity, resulting in a parameter count of 22.9 M and an inference time of 4.1 s, both of which are substantially higher than those of CFRL. This to some extent limits the deployment of FastSAM-GAL in real-time application scenarios. Nevertheless, for tasks such as large-scale postdisaster assessment, disaster loss analysis, and long-term land surface change monitoring, the reliability of detection results is often more critical than minimal inference time. In particular, in major disaster scenarios such as floods and earthquakes, frequent false alarms or missed detections in CD results may directly compromise subsequent disaster assessment, rescue resource allocation, and recovery planning, thereby incurring greater practical costs. Therefore, despite the substantially higher model complexity of FastSAM-GAL compared with CFRL, the κ score improves from 0.718 to 0.846, indicating that FastSAM-GAL is capable of more effectively mitigating the potential risks and losses associated with missed detections and false alarms.

In summary, FastSAM-GAL demonstrates a favorable accuracy–efficiency tradeoff in accuracy-critical application scenarios. Compared with methods such as AOSG, SDIR, and MCFNet, its inference time is significantly reduced; whereas compared with highly efficient methods such as CFRL, AEKAN, and CDRL, it exhibits a clear advantage in detection performance.

F. Limitation Analysis

Although the proposed FastSAM-GAL achieves favorable performance in multimodal CD tasks, several limitations still need to be further addressed in future research.

First, the current model still relies on manual hyperparameter tuning across different scenarios, particularly for the adversarial loss weight λ. As discussed in Section V-A, the optimal value of λ varies across different datasets, depending on factors such as scene coverage, land-cover complexity, and cross-modal discrepancies. This indicates that λ needs to be empirically adjusted according to scene characteristics,

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TABLE VIII

COMPARISON OF CD PERFORMANCE AND INFERENCE EFFICIENCY ON THE DATASET #2 (BEST RESULTS ARE MARKED IN RED, SECOND-BEST RESULTS ARE MARKED IN BLUE, AND ∗ INDICATES THE DEEP-LEARNING-BASED METHODS)

*PRBCD-Net			SG-SRF			*CGSL			SDIR		
Time (s)	Params (M)	κ	Time (s)	Params (M)	κ	Time (s)	Params (M)	κ	Time (s)	Params (M)	κ
1.6	0.2	0.743	8.8	\	0.838	1.0	\	0.838	28.6	\	0.795
RIEM			LPEM			AOSG			*MCFNet		
Time (s)	Params (M)	κ	Time (s)	Params (M)	κ	Time (s)	Params (M)	κ	Time (s)	Params (M)	κ
4.6	\	0.780	3.8	\	0.819	34.3	\	0.522	27.1	88.3	0.767
*AEKAN			*CDRL			*CFRL			Ours		
Time (s)	Params (M)	κ	Time (s)	Params (M)	κ	Time (s)	Params (M)	κ	Time (s)	Params (M)	κ
1.6	\	0.831	1.5	8.8	0.819	0.2	0.2	0.718	4.1	22.9	0.846

which limits the autonomous adaptability of the model to some extent. Although this article provides corresponding parameter tuning guidelines, manual hyperparameter adjustment may still hinder the deployment efficiency of the model in complex remote sensing scenarios. Therefore, future work will further explore dynamic or adaptive loss-weight learning strategies. For example, the gradient norms or magnitude variations of different loss components can be monitored during training, and the value of λ can be automatically adjusted accordingly. In this way, different loss components can maintain a reasonable relative contribution at different training stages, thereby improving the autonomy and adaptability of the model.

Second, the introduction of the FastSAM model inevitably brings a relatively large number of parameters and increased inference latency. Although this design can effectively improve the CD accuracy of the proposed method in complex multimodal scenarios, it also limits its application potential to some extent in strictly real-time tasks or edge-device deployment scenarios. Therefore, improving the computational efficiency of the model while maintaining detection performance remains an important direction for future research. Specifically, this issue can be explored from the following aspects. First, structured pruning and low-rank decomposition can be employed to compress redundant parameters in FastSAM, thereby significantly reducing the model size and computational complexity while preserving its semantic feature representation capability. Second, parameter-sharing mechanisms can be investigated to enable the model to reuse common representations during multimodal feature extraction, thus reducing redundant computational overhead.

VI. CONCLUSION

This article proposes a novel FastSAM-GAL framework for unsupervised multimodal CD. The framework employs an adversarial learning strategy to map multimodal images into a unified image domain, effectively mitigating the comparability issues caused by imaging differences. Furthermore, to address the performance degradation observed in existing methods due to insufficient feature generalization on certain datasets, FastSAM-GAL innovatively incorporates the VFM FastSAM as a frozen encoder, forming a dual-branch feature extraction architecture together with a learnable remote sensing encoder. The two branches, respectively, capture generalized semantic features and domain-specific remote sensing features, achieving complementary advantages and providing homogeneous feature representations that are both generalizable and discriminative for subsequent CD tasks. On this basis, two key modules, PMSPA and SFCEA, were further designed to enhance the overall detection performance of FastSAM-GAL. PMSPA employs a progressive multiscale feature aggregation mechanism, effectively improving the model's perception and representation of objects at different scales in remote sensing scenes. SFCEA, on the other hand, jointly models spatial edge information and high-frequency details in the frequency domain, collaboratively enhancing the model's ability to precisely delineate change boundaries. It also introduces a learnable channel vector to achieve adaptive fusion of general semantic features and domain-specific remote sensing features. Finally, change information is effectively extracted by comparing the differences between the cross-domain aligned features. Comparative experiments on five representative multimodal datasets against 12 state-of-the-art methods demonstrate that FastSAM-GAL achieves superior change localization capabilities, with $F1$-scores consistently above 0.7, highlighting its robust detection performance and strong cross-scene generalization ability. Moreover, ablation studies and robustness analyses verify the effectiveness of each key component from multiple perspectives and demonstrate that FastSAM-GAL maintains strong stability under adverse imaging conditions, further highlighting the advantages of incorporating a VFM into the multimodal CD framework.

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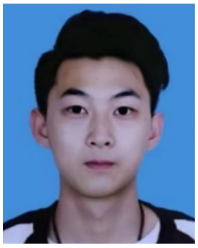
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